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Solar thermal systems and AI: Past, present, and future

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Abstract
This research explores the role of artificial intelligence (AI) in enhancing the efficiency and reliability of solar thermal, photovoltaic (PV), and hybrid energy systems. As the transition from fossil fuels becomes increasingly crucial due to their contribution to global warming and resource depletion, optimising solar energy systems through AI-driven technologies has become imperative. The study examines solar thermal and PV applications for their ability to generate electricity, heat buildings, and support industrial processes, demonstrating how AI techniques, such as artificial neural networks and machine learning models, enhance system performance and enable real-time monitoring. Additionally, hybrid energy systems, which integrate renewable and non-renewable sources like wind, solar, diesel, and fuel cells, are extensively analysed. AI applications, including support vector machines and genetic algorithms, play a key role in improving the efficiency of these systems by forecasting energy production, optimising storage, and minimising system losses. The research concludes with a SWOT analysis, identifying the strengths, weaknesses, opportunities, and threats of AI integration in energy systems while providing strategic recommendations for future research, policy development, and technological innovation. By leveraging AI in solar and hybrid energy solutions, this study offers a comprehensive framework for enhancing sustainability, reducing emissions, and ensuring a stable energy supply.

Keywords Artificial intelligence · Solar thermal systems · Solar photovoltaic systems · Energy optimisation · Hybrid energy

Abbreviations

ACO	Ant colony optimisation
AD	Anomaly detection
AI	Artificial intelligence
ANN	Artificial neural networks
ANFIS	Adaptive neuro-fuzzy inference systems
CPC	Compound parabolic concentrator
CNN	Convolutional neural networks
CSP	Concentrated solar power
DA	Direct absorption
DBNs	Deep belief networks
DL	Deep learning
DNN	Deep neural networks
DTs	Decision trees
DQN	Deep Q-Networks
ET	Evacuated tube
FL	Fuzzy logic
FOA	Falcon optimisation algorithm
FNN	Feedforward neural networks
FP	Flat plate
GANs	Generative adversarial networks
GAs	Genetic algorithms
GCP	Granular carbon powder
GO	Graphene oxide
GRU	Gated recurrent
GWO	Grey wolf optimiser
LSTM	Long short-term memory networks
MDP	Markov decision processes
ML	Machine learning
MLP	Multi-layer perceptron
MLR	Multiple linear regression
MOO	Multi-objective optimisation
MOPSO	Multi-objective particle swarm optimisation
MPPT	Maximum power point tracking
MPC	Model predictive control
P&O	Perturb and observe
PCM	Phase change material
PSC	Partial shading conditions

Extended author information available on the last page of the article

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PSO	Particle swarm optimisation
PV	Photovoltaic
RBMs	Restricted Boltzmann machines
RBFN	Radial basis function networks
RL	Reinforcement learning
RNN	Recurrent neural networks
SAHs	Solar air heaters
SCs	Solar collectors
SL	Supervised learning
SLP	Single-layer perceptron
STSs	Solar thermal systems
SVMs	Support vector machines
SVR	Support vector regression
SWHs	Solar water heaters
TES	Thermal energy storage
USL	Unsupervised learning
WNN	Wavelet neural networks

Introduction

The economic return of energy has significant potential for the economic development of the country [1]. Additionally, economic growth and increasing prosperity are closely linked to the consumption of energy resources [2]. However, increasing energy production to meet growing demand is not always the best solution. Data show that energy production often still relies on burning fossil fuels because renewable energy sources are not yet cost-competitive with fossil fuels [3]. Moreover, it is anticipated that renewable energy will take time to constitute a larger portion of overall energy usage. In other words, renewable energy sources currently need to be expanded to meet the growing energy demand. On the other hand, the consumption of fossil fuels directly triggers environmental problems such as global warming, the increase in greenhouse gas emissions, and the depletion of the ozone layer. Additionally, fossil fuels are a limited energy source since they are inherently non-renewable. Hence, depleting the energy reserves of future generations is unacceptable in terms of energy justice. Therefore, considering the environmental damage caused using fossil fuels, the necessity of moving away from this energy source is inevitable.

The way to move away from fossil fuels is to turn to alternative energy sources. Among these, solar energy is considered the most promising solution [4]. Solar energy, harnessed by absorbing sunlight and heat, is a key renewable energy source used in many applications. This energy is abundant because the sun emits large amounts of energy every day [5]. Unlike fossil fuels, it is inexhaustible and can be used as long as the sun shines. This means solar energy is virtually an unlimited resource. In summary, one of the most important advantages of solar energy is its potential to significantly reduce greenhouse gas emissions [6]. On the other hand, traditional energy sources such as coal, oil, and natural gas emit large amounts of carbon dioxide and other pollutants into the atmosphere. These emissions contribute to global warming and its harmful effects. In contrast, solar energy does not emit greenhouse gases when generating electricity. Solar energy is becoming more attractive as governments, industries, and individuals recognise the need for sustainable energy. The potential to combat climate change, reduce dependence on fossil fuels, and build a cleaner energy system makes solar energy a critical candidate for playing a key role in the future.

A robust solar energy system integrates three essential elements: solar thermal technologies, PV systems, and energy storage solutions. These components ensure the efficient and reliable use of solar energy [7]. Especially in recent years, great progress has been made in these areas, to the point that this progress is described as promising [8]. Among these, solar thermal systems (STSs) stand out as one of the most effective ways to utilise solar energy. STSs capture the heat of the sun and make this energy available for various purposes. These systems are especially valuable for industrial processes [9], building heating [10], and electricity generation. Nowadays, AI technologies are increasingly being used to enhance the efficiency and effectiveness of STSs [11]. AI algorithms are applied to better manage solar heat, optimise energy production, and monitor system performance in real time [12]. STSs transfer the heat obtained from the sun to a liquid, which is then used in processes such as heating or electricity generation [13]. Furthermore, AI-based forecasting models can make energy management more efficient by predicting solar radiation, weather patterns, and demand trends. This technology allows for broader applications of solar energy and reduces dependence on fossil fuels. In recent years, there have been significant improvements in the efficiency of STSs [14]. AI-supported optimisation methods enhance system performance, allowing more energy to be captured and used more effectively [15]. Thanks to advanced technologies, these systems can now capture more energy and utilise it more efficiently [16]. The use of AI offers a more cost-effective and sustainable energy solution for both buildings and industries. Additionally, integrating STSs with existing infrastructure has the potential to bring about radical changes in energy production [17]. AI-supported system integration and control mechanisms enable STSs to work more harmoniously with energy grids, resulting in more balanced and efficient energy production and distribution. The success of STSs has made them an essential part of solar energy technologies. These systems enable the efficient use of solar energy and contribute to creating a cleaner and more sustainable energy infrastructure. The role of AI in these systems makes it a critical element in positioning solar energy as the energy

source of the future. AI-powered analysis and predictions could lead to revolutionary changes in the energy sector by making STSs more sustainable and environmentally friendly. AI applications not only contribute to STSs but also play a key role in solar electricity systems, a topic that warrants further investigation.

Solar electricity systems incorporating PV technologies are of critical importance in the transition to a low-carbon energy future. These systems are necessary to meet increasing energy demands in a sustainable way by converting sunlight directly into electricity. AI technologies revolutionise these systems and provide more competent energy management by optimising performance and increasing efficiency. ML algorithms and advanced data analytics analyse large amounts of operational data to predict system performance, detect anomalies, and perform predictive maintenance. This facilitates the integration of solar electrical systems with the broader energy grid. Additionally, AI-based real-time data analysis enables smart grid management, enabling dynamic adjustments in energy distribution and regulating the supply–demand balance more precisely. Besides this, AI-powered energy storage solutions play a critical role in maximising the benefit of solar electricity systems. These technologies contribute to reducing costs by increasing energy storage efficiency. As a result, the integration of AI with solar electricity systems not only improves operational efficiencies but also paves the way for innovative applications such as smart homes and intelligent energy management systems. From now on, this synergy between AI and solar technologies will provide significant momentum in achieving more sustainable and efficient energy solutions.

This comprehensive review study stands at the forefront of solar energy research, particularly focussing on the integration of AI in enhancing solar thermal, PV, and hybrid energy systems. The originality of this research lies in its comprehensive examination of the intersection between AI and renewable energy technologies, providing a unique perspective on how AI can optimise system performance, improve energy efficiency, and facilitate real-time monitoring. The study explores existing literature on solar energy technologies, identifying key advancements and ongoing challenges. By reviewing a wide range of studies, it highlights the critical need for innovative solutions to enhance the performance of solar systems. For instance, the integration of ML techniques such as ANN, SVMs, and deep learning (DL) models is presented as a promising approach to improve predictive capabilities and operational efficiency. The research reviews recent advancements in AI applications, showcasing their potential in addressing issues like energy forecasting, system optimisation, and fault detection. Furthermore, this study distinguishes itself by examining hybrid energy applications that combine renewable and non-renewable sources, thereby offering a balanced and adaptable energy framework. The analysis of hybrid systems emphasises the necessity for efficient energy management strategies that minimise reliance on a single energy source while ensuring reliability. The role of AI in these systems is explored in depth, illustrating its capability to enhance performance and facilitate effective energy resource management. Overall, the study provides valuable insights for researchers, policymakers, and industry stakeholders, fostering a deeper understanding of the transformative potential of AI in solar energy applications. By addressing gaps in the current literature and proposing avenues for future research, this study serves as a crucial resource for advancing the integration of AI in sustainable energy solutions.

The subsequent stages of this comprehensive study include several key elements. "Brief description of solar thermal systems" section offers an in-depth overview of STSs, identifying the latest advancements through literature reviews from the past five years. In "Brief description of solar photovoltaic systems" section, a similar approach is taken for solar PV systems, allowing for a thorough examination of the improvements documented in the literature and a discussion of the latest technologies. The "The ways of using AI in solar thermal and photovoltaic systems" section delves into how AI addresses some challenges faced by solar thermal and PV systems, detailing the functions of various algorithms. Building on this, the fifth section presents a comprehensive literature review focussed on the performance optimisation of solar thermal and PV systems using AI. Following this, the sixth section examines the latest developments in performance prediction using a similar strategy. In the seventh section, the predictive modelling of AI-supported solar PV systems is explored in relation to weather data. The eighth section highlights the advancements reflected in publications from 2024 regarding AI applications in hybrid energy systems. The ninth section includes a SWOT analysis derived from this extensive compilation, addressing strengths, weaknesses, threats, and opportunities, while outlining necessary future actions. Finally, the study concludes with a results section that encapsulates the findings.

Brief description of solar thermal systems

STSs harness energy from the sun, which is an abundant, free, green energy source that poses no harm to human health [18]. Instead of converting this energy into electricity like PV systems, these systems absorb photon energy from the sun as it strikes the collector area, resulting in heating. This heat is then transferred to a fluid flowing through the system, raising its temperature. Consequently, according to this definition, increasing the collector area allows for more extensive interaction with solar energy, thereby enhancing

system performance [19]. These systems vary in type based on their applications, which range from residential to commercial and industrial settings; however, these are generally referred to as STSs [20]. These serve as a clean energy source thanks to solar power and play an effective role in reducing fossil fuel consumption. The common applications of these systems extend beyond just water heating and space heating and cooling [21]; they also include drying [22] and desalination [23]. Overall, compared to traditional methods, the environmentally friendly nature, cost-effective alternatives, and the clean and sustainable characteristics of STSs illuminate their growing popularity.

Figure 1 presents a schematic diagram that thoroughly illustrates the classification of different STSs [24]. This classification is based on the structural characteristics of the systems, the materials used, and their intended purposes. Today, STSs are widely employed across various fields, and their application areas are quite diverse. These systems not only meet heating and cooling needs but also play an effective role in processes, including desalination, which utilises solar energy to purify water. Moreover, these are frequently preferred in agricultural and industrial applications. For instance, solar-powered systems offer advantages in the dairy industry [25], as well as in agriculture and the hospitality sector [26].

Depending on the type of applications mentioned above, heat consumption alters significantly. In areas such as hot water provision and space heating, the highest heat consumption typically occurs within a low-temperature range of 30–150 °C. For low-temperature applications, suitable systems include flat plate (FP) collectors, evacuated tube (ET) collectors, solar ponds, and solar chimneys, with efficiencies ranging from 15 to 40%. In contrast, applications like industrial processes and energy production require higher temperatures above 400 °C. For example, the temperature range of a parabolic dish collector (tracking) is between 500 and 1000 °C [21]. Efficiency increases with rising temperatures for all types of systems. Therefore, solar collectors (SCs), which are fundamental components of STSs, hold significant importance; the technology of these collectors must meet the application requirements cost-effectively. SCs are generally categorised into two types: tracking collectors that follow the sun's movement and stationary collectors [27]. Stationary collectors encompass various types such as FP collectors, ET collectors, and stationary compound parabolic collectors [28]. Meanwhile, tracking collectors can include options like linear Fresnel reflectors and parabolic trough collectors that utilise single-axis tracking, as well as parabolic dish reflectors with two-axis tracking configurations [29]. Additionally, solar tracking configurations, types of absorbing surfaces, concentration ratios (i.e. the ratio of the aperture area to the absorbing area), and operational temperature ranges can also serve as classification criteria.

Solar water heaters

Solar water heaters (SWHs) are among the most widely used methods in solar systems, a trend that aligns with the increasing demand for water heating applications. These heaters utilise heat from solar energy to raise the temperature of a fluid. The essential components of these systems include a collector and a tank that plays a role in storing the heated water. The function of the collector is to absorb solar radiation, which causes the fluid flowing through the system to increase in temperature. This heated fluid can then be used for hot water in homes and can also provide some support for space heating simultaneously. SWH systems are typically categorised into two types: active and passive systems [30].

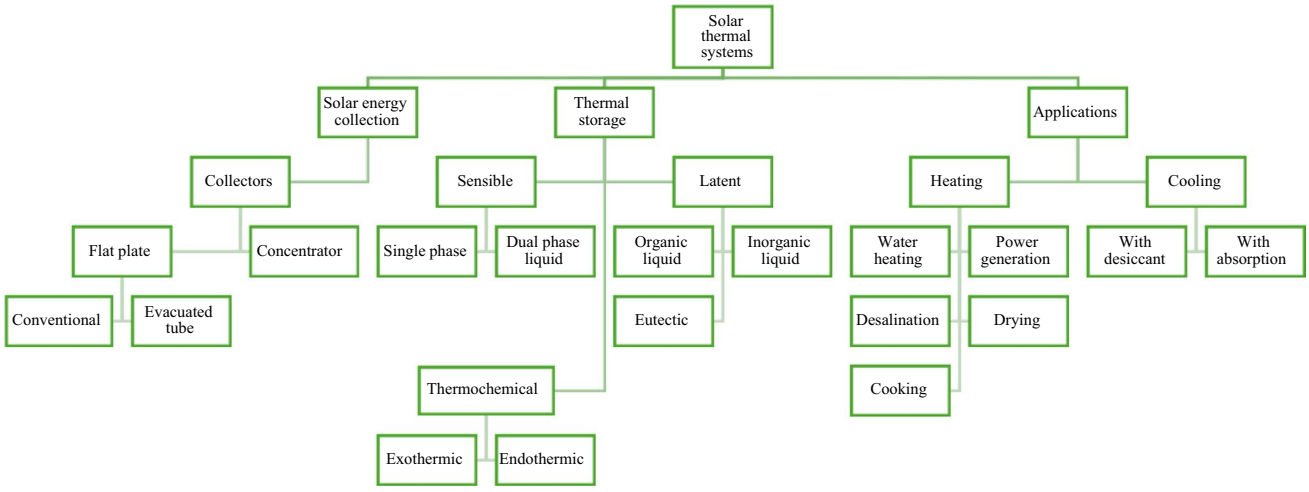

Fig. 1 Categorisation of STSs based on their design characteristics, utilised materials, and functional purposes [24]

Passive systems transfer heat through natural circulation to a tank positioned higher than the collector, typically providing hot water at temperatures between 60 and 80 °C. These systems are commonly used in residential SWHs [31]. In contrast, active systems utilise electric pumps, control devices, and valves to circulate water through the collector [32]. These systems rely on electricity to transfer thermal energy via the operation of the pump. A differential thermostat is often employed to control the water circulation when the temperature of the water in the upper header is significantly higher than that of the water in the lower tank [33]. One of the main advantages of active SWH systems compared to passive systems is their higher efficiency. Active systems achieve efficiencies between 35 and 80%, while passive systems typically range from 30 to 50% [34]. Another benefit is that active systems do not require the collector to be close to the tank, as the presence of the pump provides sufficient pressure to push the water, making these systems suitable for high-rise buildings. However, the complex structures, dependence on electricity, and the need for specialised personnel can lead to significantly higher operating costs compared to passive systems [35]. SWH systems can also be utilised in hydraulic fields, as well as in commercial and industrial applications [36]. The correct selection of these systems can reduce energy consumption, lower costs, and minimise greenhouse gas emissions. Currently, only 9% of the energy we require comes from renewable sources, but by 2040, renewable energy is expected to take the place of fossil fuels, establishing a significant position in global energy production [37]. SWH systems in homes often replace fossil fuels and electricity, providing an economical solution. These systems are suitable for low-temperature applications, offering advantages, such as simple components, low investment costs, and operational expenses. In natural circulation systems, as the water heats up, it circulates between the storage tank and the SC. When the hot water usage pattern changes, thermosiphon flow is created by solar radiation. Some thermosiphon systems employ fuzzy models to predict the water outlet temperature. Numerous SWH systems with varying designs have been studied and classified in the literature. By adjusting the angle of the SC, it is possible to capture sunlight for a more extended period. Collectors are usually painted dark to absorb more sunlight, with low-iron tempered glass placed on top [38]. The primary components of an SWH system include the SC, heat storage tank, heat transfer fluid with a heat exchanger, and heat-absorbing nanoplates, all of which play a crucial role in the system's efficiency [39], as shown in Fig. 2.

Some of the recent literature studies related to the improvement of these systems from the past five years (2020–2024) are presented in Table 1.

When reviewing the studies, it is emphasised that, since the system derives its power from solar energy, it is crucial to ensure insulation and ES solutions for conditions when solar energy is not available. Furthermore, there is a focus on utilising nanofluids to enhance system performance and selecting materials with high thermal conductivity to further increase the fluid temperature. Among the various collector types, FP collectors are known to be the most cost-effective; however, their drawback is low efficiency in cold regions. Additionally, a comparative table highlighting both the positive and negative aspects of other collector types has been created in Table 2. According to this table, ET collectors exhibit high efficiency in low-temperature and cloudy conditions, while CPCs are well-suited for high-temperature applications and can be utilised at various angles. Common drawbacks include their higher costs and the need for extra caution due to the potential for breakage in ET collectors. These advantages and disadvantages are summarised in the table below [60].

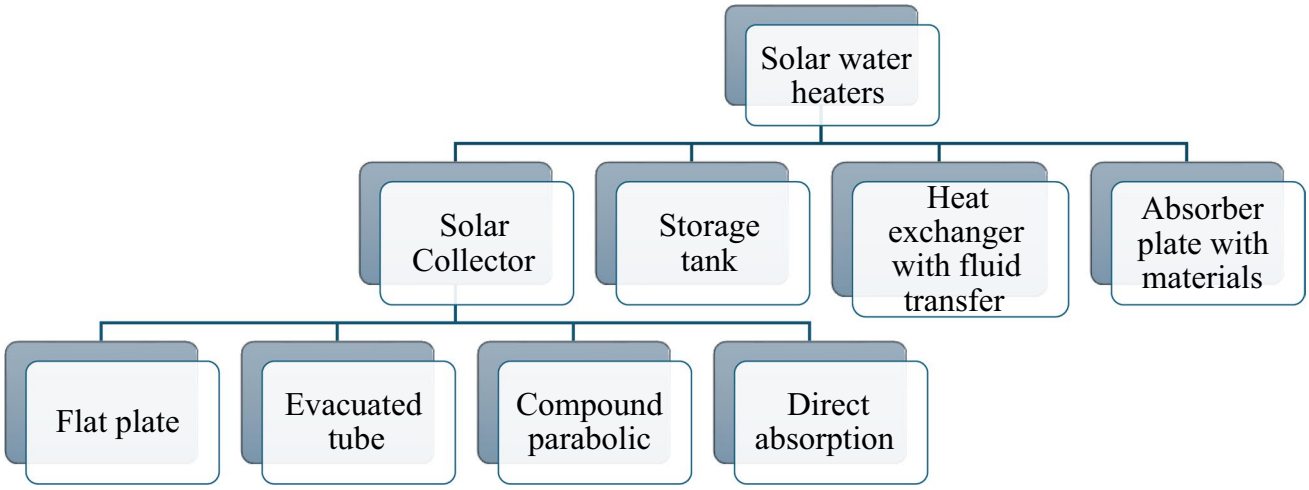

Fig. 2 The primary components of an SWH system representation

Table 1 A comprehensive review on improving the performance of SWH systems between 2020 and 2024

References	Application	Composition/Changes	Improving place	Findings
[40]	Contemporary Integrated collector storage with compound parabolic concentrator (CPC)	Transparent vacuum insulation within the annular space and automated insulation system at night	Daily efficiency as well as loss coefficient	Numerical studies show that with optimal night-time insulation and vacuum conditions, daily efficiency reaches 58.62%, and the loss coefficient drops to 1.68 W m^{-2} K^{-1}. Experimental results reveal that the integrated collector storage loss coefficient decreases from 5.41 to 3.34 W m^{-2} K^{-1}
[41]	FP	Phase change material (PCM) and fin	Collector efficiency	Results show that the system with PCM provided lower temperatures in the morning but extended hot water availability in the evening. Average efficiency increased from 33 to 46% with PCM. The addition of fins improved storage capacity, especially with high melting point PCMs, though fins may lead to greater afternoon heat loss, slightly reducing efficiency
[42]	Direct absorption (DA)	Graphene oxide (GO), zinc oxide, copper oxide, and their hybrids	Photothermal performance	The photothermal efficiency of the nanofluid containing 0.0012% GO in water is 46.6% higher than other nanofluids. This increase is attributed to the excellent stability, high absorbance, and thermal conductivity of GO nanofluid, suggesting its potential use as a direct solar energy absorber in solar thermal energy conversion applications
[43]	SWH with a galvanised iron absorber	Replacement of glass glazing with low-density polyethene (LDPE) glazing; closed-loop system with constant flow rate; integrated 4E analysis (energy, exergy, economics, efficiency recovery ratio)	Enhancing energy and exergy efficiency, reducing mass and cost, and lowering carbon footprint	LDPE glazing achieved 5.57% higher energy efficiency and 3.2% higher exergy efficiency than glass. Mass reduced by 32.56%, with 40.9% cost savings compared to a geyser. LDPE showed higher solar transmissivity (0.92 vs 0.81), quicker dynamic response in September, and longer in October. Carbon credits were valued at INR 419.46 for LDPE vs INR 125.02 for glass. The payback period was 7 months and 24 days
[44]	FP	Straight, curve, and spiral tube	Performance of system	A maximum efficiency of 73% was achieved at a flow rate of 0.006 kg s^{-1}. At 0.0045 kg s^{-1}, straight, inclined, and spiral tubes provided efficiencies of 58%, 62%, and 69%, respectively, while at 0.006 kg s^{-1}, these efficiencies increased to 62%, 65%, and 73%

Table 1 (continued)

References	Application	Composition/Changes	Improving place	Findings
[45]	ET	Water-in-glass ET Tested under the Kenyan climate Daily temp & efficiency measured	Daily efficiency and final outlet temperature	The results show that daily efficiency ranged between 58 and 65%, and with an initial temperature of 25 °C, the final daily outlet temperature exceeded 55 °C
[46]	Non-tracking CPC	Three different volumetric concentrations (0.025%, 0.05%, and 0.075%—Flowrates of 0.01 kg s^{-1}, 0.02 kg s^{-1}, 0.05 kg s^{-1}, and 0.07 kg s^{-1}—with Al_2O_3 − water and only water fluid	Heat gain and system performance improvements	The maximum system performance with 0.075% Al_2O_3 nanofluid was achieved during peak solar radiation, with heat gains of 18.58% at low temperatures and 21.22% at high temperatures. The average thermal efficiency increase was 13.5%, and the use of nanofluid improved system performance by 43% compared to water
[47]	DA	Flow rate 2–10 L min^{-1} Nanoparticle concentrations 0.0015–0.082mass% With carbon-based nanofluid	Thermal efficiency	At a particle concentration of 0.01%, the collector achieves an average thermal efficiency of 80%. For comparable temperature differences, the DASC shows a 5.8% to 37.9% higher efficiency than a collector with similar geometry but surface absorption of light energy
[48]	Vacuum FP	Curved evacuated cover and rear Rockwool insulation	Thermal efficiency	The vacuum FPSC demonstrates higher thermal efficiency than FP and ET ones by up to practically 28%, with reduced heat loss and lower cost, making it a promising solution for hot water production
[49]	U-Pipe ET	Without PCM PCM and silicone oil Nanoboosted PCM	Thermal performance	The U-Pipe ETC exhibited higher peak water temperatures and faster heat transfer compared to the heat pipe ETC, though with a shorter useful hot water duration. The incorporation of nanoparticles enhanced energy storage, extending the useful duration by more effectively utilising latent heat storage
[50]	Non-tracking CPC	Multi-walled carbon nanofluid-SiO_2 + Ethylene glycol–water hybrid nanofluid with various volumes	Thermal performance	Hybrid nanofluids substantially boost thermal efficiency, achieving a peak increase of 14.27% over the base fluid, and also enhancing exergy efficiency. Despite a 9.72% rise in pumping power, this increase remains negligible relative to the net useful heat production, demonstrating robust overall thermal performance

Table 1 (continued)

References	Application	Composition/Changes	Improving place	Findings
[51]	Tubular DA	Multi − walled carbon nanofluid + Fe_3O_4 with volumetric concentration in 0.0045–0.0053%	Thermal efficiency	This study reveals that the thermal efficiency of the tubular DA collector with Fe_3O_4/MWCNT nanofluid is slightly higher than that of surface absorption collectors, ranging from 52.3 to 69.4%. Additionally, the MWCNT nanofluid outperforms the hybrid nanofluid in terms of thermal efficiency due to its higher spectrum-averaged extinction coefficient
[52]	FP	Straight, wavy, and helical models	Thermal performance boost	The straight pipe model demonstrated a thermal performance that is 12.3% higher than that of other models
[53]	ET	PCM + MF	Collector output temperature, ES efficiency, and overall thermal efficiency	The self-storage collector exhibited better performance with an ES efficiency of 75% and an overall thermal efficiency of 72%, compared to the normal collector's values of 57% and 49%, respectively. Additionally, at the end of the experiment, the self-storage collector-maintained water at 63 °C, while the normal collector remained at 17 °C
[54]	Involute-shaped CPC	Collector angle variation Concentration ratio alteration Orientation	Thermal performance	The CPC SC with a variable concentration ratio showed 16% better optical performance and 7.3% higher thermal performance in the E–W orientation compared to the N–S orientation. Additionally, a maximum efficiency of 70% was achieved, highlighting the need for a solar tracking system for more efficient operation
[55]	DA	Different climates in Iraq: Baghdad, Samawa, Mosul, and Al-Qa'im	Thermal energy and exergy efficiency	SCs in Samawah demonstrate the highest thermal energy efficiency, reaching up to 66.5%, while Al-Qa'im records the highest exergy efficiency at 36.21%. Environmentally, the collector in Al-Qa'im achieves significant CO2 reduction, mitigating 2.73 kg m^{-2} of collector annually

Table 1 (continued)

References	Application	Composition/Changes	Improving place	Findings
[56]	Stainless-steel FP	Severe width of the microchannel and the height of the corrugations	Thermal performance	This study demonstrated a significant thermal performance increase with the stainless-steel FPSC developed through microchannel stamping, achieving an instantaneous efficiency of 86.10%. Additionally, the design improved temperature uniformity and durability, while the mathematical model clearly illustrated the relationship between environmental conditions and collector efficiency
[57]	ET	Copper and aluminium using fin material, the inclination angle on ETSWH, severe mass flow rate (2–5 l/h)	Thermal efficiency	At a flow rate of 2 l/h, the highest outlet temperature is recorded at 55 °C, while at 5 l/h, the highest temperature is 41.2 °C. Copper fins are consistently more efficient than aluminium fins due to their superior thermal conductivity. Additionally, the efficiency of the ETSWH is found to be optimal at a 30° inclination angle, with the copper fin model achieving an efficiency of 58.57%, compared to 51.50% for the aluminium fin model
[58]	CPC	Aerogel and polycarbonate cover on the system	Collector efficiency	The results indicate that the polycarbonate coating reduces the optical performance by approximately 16% while protecting the reflector from environmental factors. Additionally, the aerogel-equipped CPC was observed to increase collector efficiency by 30.6% under conditions of 60 °C inlet water temperature, 20 °C ambient temperature, and 700 W m^{-2} solar radiation
[59]	DA	Carbon black nanofluid with 0.05 mass% Alteration of flow rates and tilt angles	System efficiency	The efficiency of the DASC increased with the rise in flow rate. The highest efficiency was achieved when the surface was positioned facing downwards (0° tilt angle), providing a 35% higher efficiency compared to when the surface faced upwards (180° tilt angle). CFD model simulations indicated that the optimal efficiency occurred at a concentration of 0.05 wt.% carbon black

Table 2 Positive and handicap features in accordance with the collector types [60]

Type	Pros	Cons
FP	Cost-effective Simple design Robust performance Versatile application Low maintenance requirements	Lower efficiency in cooler climates Higher heat loss Larger size needed for high output Limited performance in low light conditions
ET	High efficiency in low temperatures and cloudy conditions Excellent insulation Better performance in winter Reduced maintenance due to fewer moving parts	Higher cost Fragile tubes Can be visually intrusive Potential for reduced performance if tubes become dirty
CPC	High efficiency through sunlight concentration Reduced heat loss Better performance at various angles Can achieve higher temperatures	Complex design Performance sensitive to orientation Higher initial cost Requires precise tracking or orientation adjustments
DA	High efficiency with DA Reduced heat loss Can handle higher temperatures Efficient use of nanofluids for improved heat transfer	Complex design and operation Dependence on costly nanofluids Requires specialised components Potential for higher maintenance due to fluid handling

Solar air heaters

Solar air heaters (SAHs) are a cost-effective solution for providing heat in homes, offices, and industries [61]. It has low design and maintenance costs [62]. However, air has poor thermal conductivity, which leads to weak heat transfer between the air and the absorber plate [63]. As a result, the absorber plate gets very hot, giving rise to significant heat loss to the surroundings [64]. This reduces the overall efficiency of SAH systems. The poor heat transfer is also caused by the formation of a viscous sub-layer, which brings about low thermal performance [65]. To improve the collector's efficiency, studies have explored various techniques that increase heat transfer by creating turbulence and expanding the surface area. These methods include adding fins [66], ribs [67], baffles [68], and porous materials like wire mesh [69], limestone [70], iron scraps [71], and gravel [72] inside the collector. Different airflow types, such as double-pass counterflow [73], counterflow [74], and multiple air passes [75], are also used to enhance heat transfer and improve heat extraction. Many experimental and analytical studies have been conducted to evaluate the thermal performance of SAHs. These studies cover various designs of SAHs with different sizes, airflow types, measuring tools, and testing conditions, both indoors and outdoors, under static and dynamic setups. The range, use, and accuracy of these instruments are also discussed. The literature highlights the importance of duct modifications and energy storage devices in improving thermal performance. Additionally, it outlines various techniques for enhancing performance, including the best values for geometrical and flow parameters. Researchers have also reported key performance indicators like thermal, thermo-hydraulic, and exergy efficiencies, showing their impact on thermal performance improvement. Table 3 includes only a few articles from the last 5 years where thermal performance improvement has been investigated.

Solar concentrators

Solar concentrators are devices that focus sunlight onto a smaller area, increasing the energy captured from the sun [99]. This concentration of solar energy boosts the efficiency of energy conversion, making these systems important for generating electricity, industrial heating, solar cooling and solar cooking, and desalination [100, 101]. The main idea behind a solar concentrator is to collect sunlight over a large surface and direct it onto a receiver [102]. Here, the concentrated energy is either turned into heat or electricity [103]. These systems can reach very high temperatures, making them useful for both small and large-scale energy applications.

There are various types of solar concentrators, each optimised for specific applications. Parabolic troughs, for instance, use a curved reflective surface to direct sunlight onto a long receiver tube, where a fluid is heated to high temperatures [104]. This fluid is then utilised to generate steam, which drives turbines in concentrated solar power (CSP) plants for electricity production [105]. On the other hand, parabolic dishes concentrate sunlight onto a single focal point, achieving much higher temperatures, making them ideal for small-scale systems like Stirling engines [106]. Fresnel reflectors offer a more compact and cost-effective design by using flat mirrors to direct sunlight onto a receiver, although they tend to be slightly less efficient than parabolic systems [107]. Solar power towers adopt a more sophisticated approach, utilising a field of heliostats to reflect sunlight onto a receiver located at the top of a tower [108]. This design is particularly efficient, especially when

Table 3 A comprehensive review on improving the performance of SAH systems for the last 5 years of studies

Reference	Duck type	Cross Section	Novel Configuration	Efficiency value
[76]	Hybrid duct with parallel pass airflow paths	Upper side: Rectangular Lower side: Triangular	Roughened with inclined shape ribs	80.1% (with roughness pitch 8, height 0.021, angle 0.5) 22.4% compared to conventional rectangular duct SAH 18.1% compared to conventional rectangular duct SAH
[77]	Conventional	Rectangular	No thermal energy storage (TES) integration, standard design	39.22%
	Modified	Cylindrical	Copper tube with granular carbon powder (GCP) as TES	56.01%
	Modified	Cylindrical	Copper tube with paraffin wax as TES	69.61%
	Modified	Cylindrical	Copper tube with a combination of GCP and paraffin wax as TES	78.31%
[78]	Single pass	Triangular	Absorber plate with 1% graphene nanomaterial in black paint coating	48.23%
			Absorber plate with standard black paint coating	43.15%
[79]	SAH-A	Standard	–	~50%
	SAH-B	Helical tubes	Paraffin Wax	~57.41%
	SAH-C	Helical tubes	Paraffin Wax + GCP	79.10%
[80]	Double pass	V-groove	Absorber plate with pin-fins on both sides	Thermal Efficiency: 17.4% improvement over conventional Heat Transfer Coefficient: 61.6 W m^{-2} K^{-1} Mean Outlet Temperature: 45–55 °C
[81]	Double pass	Triangular	Triangular absorber plate	Ultimate Daily Thermal Efficiency: 63.77% Instantaneous Efficiency: 30–80% (0.0043 to 0.0125 kg s^{-1})
		Metal Fibre	Metal fibre as absorber surface	Ultimate Daily Thermal Efficiency: 74.2% Instantaneous Efficiency: 30–95% (0.0043 to 0.0125 kg s^{-1})
[82]	Double pass	Rectangular	3 waves fin	Thermal efficiency: 80%
			7 waves fin	Thermal efficiency: 84%
[83]	Cylindrical	V-Type Corrugated	Without Using PCM	For 0.006 kg s^{-1}: Thermal efficiency 44.8% For 0.01 kg s^{-1}: Thermal efficiency 53.1%
			With PCM 40	For 0.006 kg s^{-1}: Thermal efficiency 50.1% For 0.01 kg s^{-1}: Thermal efficiency 58.2%
			With PCM 50	For 0.006 kg s^{-1}: Thermal efficiency 51.5% For 0.01 kg s^{-1}: Thermal efficiency 62.6%
			With PCM 40&50	For 0.006 kg s^{-1}: Thermal efficiency 53.39% For 0.01 kg s^{-1}: Thermal efficiency 61.4%

Table 3 (continued)

Reference	Duck type	Cross Section	Novel Configuration	Efficiency value
			With PCM 4 ∗ 40&5 ∗ 50	For 0.006 kg s^{-1}: Thermal efficiency 50.4% For 0.01 kg s^{-1}: Thermal efficiency 60.7%
[84]	Jet-impingement SAH	Duck's aspect ratio: 12	Transverse Ribs on Absorber Plate	Optimal thermal efficiency for 0.039 kg s^{-1}: 78%
[85]	Triangular	Triangular Flow Passage	Hemispherical Dimples on Heat-Collector Plate	Nusselt Number: 5.33, Thermohydraulic Performance: 3.48 Enhanced thermal performance: 38.42% compared to the traditional one
[86]	Multi-geometry baffle	Rectangles, Slits, Cylinders	Multiple geometry arrangement using rectangular protrusions, slits, and cylinders	The average thermal efficiencies for air flow rates of 0.01, 0.02, 0.03, and 0.04 kg s^{-1} are 40.8%, 58.2%, 68.2%, and 77.4%, respectively. The maximum efficiency was recorded at 0.04 kg s^{-1} air flow rate, reaching 92.2%
[87]	Louvred fins	Louvre angle: 20°, length: 2.5 cm, pitch: 2.5 cm, height: 3.5 cm	Louvred fins fitted to the absorbing plate	The maximum efficiency was 70% at an air flow rate of 0.0158 kg s^{-1}, representing a 106.7% increase compared to the flat SAH using 2 cm spaced louvred fins
[88]	Arc ribbed wire	Wire diameter: 2 mm (from 0 to 0.5 m), 2.5 mm (between 0.5 and 2 m)	Variable spring wire arrangement	The maximum thermal efficiency is 52%, the effective efficiency is 51.4%, and the exergy efficiency is 2.2%
[89]	Double-pass SAH with corrugated absorber plate and curved reflector	Double-pass, external reflector, and corrugated absorbing plate arrangement	Curved reflector usage	The maximum thermal efficiency increase is 19.33%, the thermohydraulic efficiency increase is 19.84%, and the exergy efficiency is 1.14%
[90]	Double-pass SAH with spiral and straight tubes filled with PCM	Double-pass, spiral, and straight tube structure	Use of copper tubes filled with paraffin wax and glycerol for energy storage purposes	In the spiral tube configuration, the thermal efficiency is 29.86%, the exergy efficiency is 1.11%, and the maximum air temperature reached is 59 °C
[91]	Modified	Cylindrical Containers	Coconut Oil PCM	Thermal efficiency: 71.35% for forced convection
			Paraffin Wax PCM	Thermal efficiency: 75.7% for forced convection
[92]	Frustum Roughened	Frustum Shape	Roughened SAH with Various Roughness Parameters	Maximum Thermal Efficiency: 79%
[93]	Asphalt SAH Perforated Metallic SAH Combined SAH	Rectangular Perforated Plate Combined Asphalt and Perforated	Asphalt pavement and perforated metallic collector usage	The thermal performance of the combined SAH reaches approximately 60% in the morning and close to 80% in the afternoon
[94]	Recycled Materials	Cylindrical	Waste Beverage Cans + Recycled Plywood	Energy efficiency: ~ 21% Exergy efficiency: 1.8%

Table 3 (continued)

Reference	Duck type	Cross Section	Novel Configuration	Efficiency value
[95]	Roughened Plate SAH with Inclined and Winglet Baffles	Inclined and Winglet Baffles	Incorporates inclined and winglet baffles for artificial roughness	The adjusted system achieves the peak daily efficiency of 85.1% and the highest average thermal performance of 72.9%
[96]	Continuous and Discrete Finned Absorber SAH	Rectangular Tubes and V-Corrugated Fins	Hybrid design combining continuous rectangular tubes with V-corrugated fins	Efficiency: Peak energy efficiency of 79% and 82.5% for continuous and discrete fin structures at Re 10,930, respectively; Heat transfer coefficients improved by 87.2% and 59.3% over plain absorbers
[66]	Hollow Rectangular Fins with PCM	Fins with PCM dimensions of 0.06 m × 0.015 m, 0.06 m × 0.025 m, and 0.09 m × 0.015 m	Zigzag Fins with PCM	*Energy efficiency* 0.06 m × 0.015 m: 48.1% 0.06 m × 0.025 m: 59.9% 0.09 m × 0.015 m: 65.7%
[68]	Longitudinal Baffles Transverse Baffles	N/A	Investigate optimum baffle numbers and types	At a mass flow rate of 0.026 kg s^{-1}, N(baffle number) = 4 recorded the peak effective efficiency of 54.32%; for mass flow rates of 0.039 kg s^{-1}, 0.052 kg s^{-1}, 0.065 kg s^{-1}, and 0.078 kg s^{-1}, N = 6 exhibited the highest effective efficiency
[97]	Triangular	Transformative cross-sectional shape akin to a triangle	Cinque-faced rib-roughened triangular channel SAH	Thermal efficiency: 80.7%
[98]	SAH	FP + ET	Integration of FP and ET collectors with paraffin C20-C33 as PCM	Maximum daily thermal efficiency: 64.53% (at 0.025 kg s^{-1} with PCM)

combined with thermal ES, allowing for electricity generation even after sunset [109]. Lastly, CPCs provide a simpler solution by capturing sunlight from a broad range of angles, making them effective in diffuse light conditions without requiring precise sun-tracking [110].

While solar concentrators offer numerous advantages, including high efficiency and the ability to generate significant heat, they also face several challenges. For optimal performance, most systems require precise sun-tracking, which can be enhanced by incorporating AI [111]. AI-driven tracking systems can predict the sun's movement and adjust concentrators in real time, maximising energy capture and improving efficiency [112]. However, large-scale installations often demand substantial land areas, and the initial investment costs can be high, particularly for advanced technologies like solar power towers. Despite these obstacles, solar concentrators remain versatile, finding applications beyond CSP, such as in industrial heating, solar cooking, solar stills, and thermal ES. By storing heat in materials like molten salts, these systems ensure a continuous energy supply, even during cloudy conditions. Consequently, solar concentrators represent a promising solution for both small decentralised applications and large-scale power generation.

Brief description of solar photovoltaic systems

Solar energy is considered one of the most promising and clean sources of renewable energy, with many applications all around the world. Because of its widespread usage, solar PV technology has become especially important. This is due to its incredible versatility and flexibility in design [113].

As the global demand for energy continues to grow, driven by rising consumption, population increases, and economic development, renewable energy sources like solar PV are gaining more and more attention [114]. This is happening because the reserves of fossil fuels, which we have relied on for so long, are gradually running out [115]. Against this, solar PV systems provide a sustainable solution that can meet these energy demands, all while reducing negative impacts on the environment, such as global warming and ozone depletion [116].

In recent years, the solar PV market has grown at an astonishing rate. For instance, China has been at the forefront of solar PV adoption. The total energy generated from solar PV systems in China has reached a massive 400 GW, far surpassing many other countries, including developed nations [117]. It is predicted that this increment could easily exceed 800 GW by 2024 and 4000 GW by 2032 [118]. However, other countries, such as the USA and India, are expected to generate around 100 GW of energy from solar PV systems by 2023, staying well below China's achievements [119]. Meanwhile, off-grid solar PV systems are becoming an important solution in developing regions, such as sub-Saharan Africa and parts of Asia, where they are helping to bring electricity to areas that previously had no access [120].

The basic principle behind solar PV technology is the conversion of sunlight into electricity using semiconductors, which are typically organised into solar cells [121]. When sunlight, in the form of photons, hits these cells, it creates an electrical current. One of the main advantages of solar cells is their long lifespan, with some able to work efficiently for more than 30 years, both on Earth and in space [122].

To understand how PV systems work, a common model used is the one-diode model [123]. This model is simple but effective, allowing engineers to accurately predict how a solar PV system will behave under different conditions [124]. Various models, such as the one-diode (simple) and two-diode (complex) models, are useful for understanding the current–voltage (I–V) and power–voltage (P–V) characteristics of a PV system [125]. However, despite all these advantages, solar PV systems are not without their challenges. Because they are installed outdoors, they are constantly exposed to physical and environmental stress [126]. Over time, this can cause damage, such as cracks in the solar cells, corrosion, or even the delamination of the cells [127]. Another issue is shading, where part of the solar PV array is blocked by shadows. This can cause a mismatch in the I–V characteristics of the system, leading to localised heating, which may damage the cells like fires [128]. Electrical faults, such as loose connections or poor soldering, can also reduce the efficiency of the system, leading to a drop in the amount of energy it can produce [126].

In recent years, AI has emerged as a key tool for improving the performance of solar PV systems, especially when it comes to identifying faults early [129]. AI-based algorithms can process the large amounts of data generated by PV systems to find problems and predict potential failures before they result in a significant loss of energy production [130]. For example, AI-powered fault detection systems can detect microcracks in cells or recognise shading patterns that are reducing energy output [131]. This allows for quick interventions, preventing more serious damage. In addition to fault detection, AI technologies can also help automate the maintenance of solar PV systems [132]. This reduces the need for human intervention and ensures that the system is operating at maximum efficiency. AI can even provide real-time predictions based on changing environmental conditions, helping to optimise energy production [133]. These advances in AI not only extend the operational life of solar PV systems but also make them more reliable and efficient, ensuring that solar energy keeps playing a vital critical role in the future of renewable energy.

The ways of using AI in solar thermal and photovoltaic systems

AI plays a crucial role in enhancing the efficiency, optimisation, and control of STSs [134]. One key application is in predictive modelling, where AI algorithms, such as ML and DL, are used to forecast solar radiation and weather conditions, helping to optimise the system's performance [135]. AI can also be employed in real-time monitoring and fault detection, allowing systems to operate at peak efficiency and promptly identify issues [136]. In SWHs and SAHs, AI can optimise flow rates, temperature regulation, and heat storage by dynamically adjusting system parameters based on environmental conditions. Moreover, AI-driven control systems improve the energy output of solar concentrators by adjusting the angles of mirrors or lenses for maximum sunlight capture [137]. Through the integration of AI, STSs can become more adaptive and intelligent, reducing energy wastage and improving sustainability.

AI-based systems learn and train continuously in order to behave like humans and develop self-reasoning and problem-solving capabilities [138]. Properly trained systems using the AI approach could solve complex problems smoothly without a higher degree of mathematical manipulations. Therefore, the AI approach has become an alternative and promising approach to various conventional problem-solving techniques. AI consists of various techniques, namely ML, DL, ANN, anomaly detection (AD), decision trees (DTs), GAs, reinforcement learning (RL), fuzzy logic (FL), recurrent neural networks (RNN), long short-term memory networks (LSTM), SVM, Q-learning, and their various hybrid approaches.

Figures 3 and 4 illustrate the applications of AI in different sections of STSs and solar PV systems.

Deep learning

ML tasks are generally categorised into three main types: supervised, unsupervised, and semi-supervised learning [139]. In supervised learning (SL), the training of the model involves an optimisation process aimed at minimising the error by comparing the predicted outputs with the original data labels [140]. Unsupervised learning (USL), on the other hand, explores the internal structure and hidden relationships within a dataset using unlabelled data, often applied in dimensionality reduction, clustering, and generative models [141]. Semi-supervised learning is a technique that combines labelled and unlabelled data for model training. In situations where labelled data are scarce, this method enhances accuracy and provides efficient results [142]. In this context, a general review of algorithms involving DL will also be presented in the subsequent sections.

Artificial neural networks

ANN is a model inspired by biological neural systems, which are highly effective in understanding complex relationships within datasets [143]. Unlike traditional methods, ANN can learn relationships among complex data without relying solely on mathematical models, potentially offering computational cost advantages. These models process data through layers, where connections at each layer are defined by adjustable parameters such as masses and biases [135]. ANN is typically employed within an SL framework and yields successful results in tasks such as classification and regression [144]. A basic ANN model consists of an input layer, one or more hidden layers, and an output layer [145]. Information from the input layer is processed in the hidden layers, which extract data features through nonlinear activation functions. During this process, activation functions, such as logistic sigmoid functions, are often employed to constrain neuron outputs between 0 and 1 [146]. Single-layer perceptron (SLP) models are among the simplest ANN

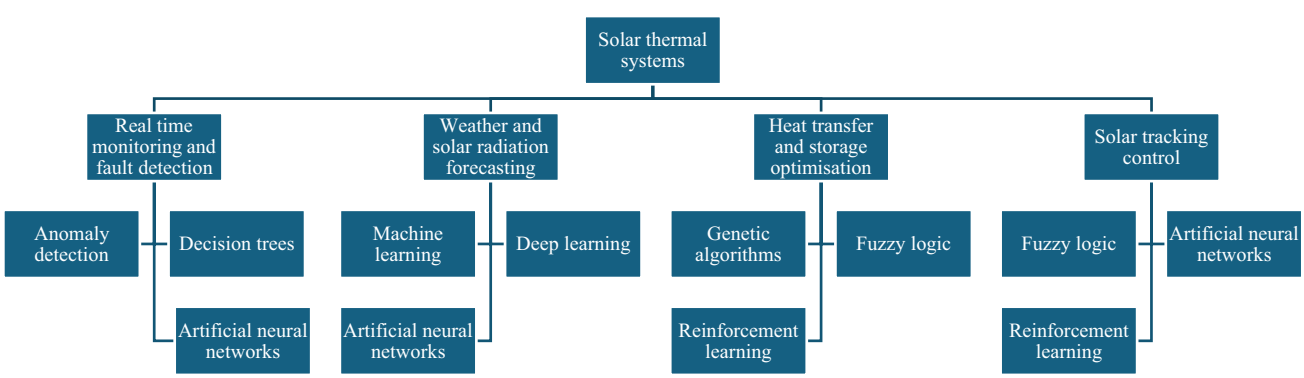

Fig. 3 Various classifications of AI in accordance with the applications in STSs

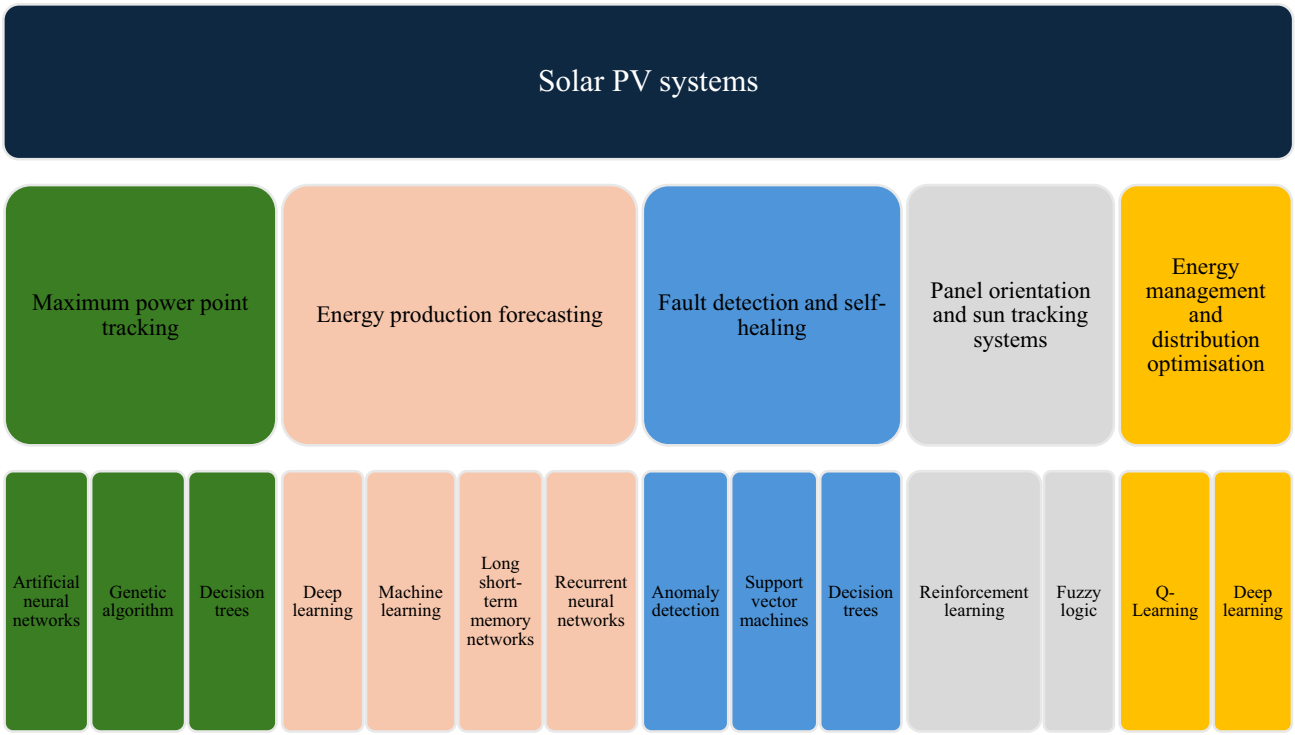

Fig. 4 Various classifications of AI in accordance with the applications in solar PV systems

structures, commonly used for basic, linear classification problems [147]. While SLPs exhibit a feedforward structure for binary classification tasks, multi-layer perceptron (MLP) provides solutions for more complex, nonlinear classification and regression issues [148]. MLPs contain multiple hidden layers between input and output layers, and the connections among these layers are continuously updated through the backpropagation algorithm [149]. One of the MLP's strengths is its high performance in nonlinear and complex problems. However, an increase in the number of hidden layers can lead to issues such as overfitting, necessitating careful structuring [150]. The success of this model depends on the correct arrangement of relationships among layers; thus, selecting appropriate hyperparameters during the training process is critical. ANN stand out as a significant optimisation tool, particularly in solar energy applications. In STSs and solar PV systems, ANN models are frequently employed to enhance system performance and optimise energy efficiency. Commonly used ANN models in PV systems include feedforward neural networks (FNN), radial basis function networks (RBFN), RNN, convolutional neural networks (CNN), and LSTM. Each of these models possesses unique characteristics and application areas, offering the potential to optimise energy production. Detailed literature summaries of these models will be presented in subsequent sections. In conclusion, ANN are widely used in both solar thermal and PV systems. These models are regarded as effective tools for optimising energy production and management processes, facilitating the increased adoption of AI technologies in solar energy applications.

Recurrent neural networks

These networks possess the ability to process time-related datasets, creating an intrinsic memory mechanism by evaluating connections between sequential data. Unlike traditional neural networks, these structures utilise sequential information to relate each step to previous computations, ensuring that the influence of past data is not lost [151]. However, when working with large data sequences, RNNs can experience performance degradation due to a tendency to forget past information. To address this limitation, the LSTM model has been developed, enabling the retention of data for longer periods and thus facilitating more accurate results [152, 153]. A key advantage of LSTM networks is their capacity to store long-term information through memory blocks. The functioning of these memory blocks is managed by mechanisms involving the input gate, forget gate, and output gate, which regulate the entry and exit of data [154]. The input gate determines which information will be processed, while the forget gate decides whether previous information should be retained, and the output gate controls which data will be outputted. This architecture allows LSTMs to deliver enhanced performance compared to traditional RNNs,

making more accurate predictions over longer-term relationships. The success of RNNs in sequential data analysis is particularly evident in areas such as speech recognition, machine translation, and image captioning [155]. While traditional neural networks operate under the assumption that all inputs are independent, RNNs leverage historical information to produce more coherent predictions. The fundamental principle of RNNs involves combining the previous hidden state with the current data, allowing them to function like a memory in future predictions. This makes RNNs particularly suitable for applications requiring sequential data analysis. In energy forecasting, RNNs and their derivatives are frequently employed, providing significant advantages in understanding the sequential structure of data. The LSTM model, in particular, allows for more precise results in energy production forecasting by effectively processing long-term dependencies [156, 157]. Such structures play a vital role in enhancing accuracy in fields requiring the analysis of sequential data, such as energy production.

Long short-term memory recurrent neural network

LSTM networks stand out as structures equipped with memory cells specifically designed to address the vanishing gradient problem; these cells replace the nodes found in traditional RNN [158]. The fundamental architecture of LSTM operates through the interactions of gates within the memory cell. These gates manage the information entering and exiting the memory cell: the forget gate determines which information should be discarded, while the input gate selects which parts of the incoming data will be updated in the memory cell. Simultaneously, the update gate creates new cell values, allowing for dynamic management of the cell state [159]. With this configuration, LSTM can effectively learn long-term dependencies. The development of LSTM was first introduced by Sepp Hochreiter and Jürgen Schmidhuber, and it has been observed to yield successful results in time series forecasting problems [160]. The advantages of LSTM include critical roles in predicting power output for medium- and long-term solar PV systems and improving the performance of STSs [156, 161]. As an alternative to LSTM, the gated recurrent unit (GRU) features a more simplified architecture. The GRU combines the internal state with the hidden state, merging the forget and input gates into a single gate [162]. Research has indicated that GRUs perform better in long-term forecasting, while LSTMs excel in short-term predictions; however, both structures offer significant advantages compared to standard RNNs [163]. Looking forward, the integration of LSTM and similar AI methods with more complex datasets holds the potential to enhance the overall effectiveness of the model. With ongoing developments in DL, LSTMs may expand into new application areas. In conclusion, the importance of the LSTM structure in time series forecasting and complex data analysis is substantial, and a thorough understanding of its operation plays a critical role in enhancing the model's effectiveness.

Gated recurrent unit

GRU, introduced by Cho et al. [164], is a popular variation of RNN. As GRU operates on principles similar to LSTM, it can be regarded as a derivative of LSTM. Both models employ a closed mechanism to regulate the flow of information. However, in GRU, the input and forget gates are combined to create a single update gate. Unlike LSTM, GRU features only two gates: the update gate and the reset gate. These gates determine which pieces of past information should be retained and which should be discarded. A notable advantage of GRU is its simpler structure compared to LSTM, as it contains fewer parameters [165]. This characteristic makes GRU more efficient and compact. Additionally, the training time for GRU is faster than that of LSTM [166]. The update gate, one of the two gates in the model, adjusts how much of the previous information is retained, while the reset gate controls which information will be disregarded. Therefore, GRU can significantly reduce training time without compromising prediction performance [167].

Convolutional neural networks

CNNs have become popular in the field of DL due to their high success rates and accuracy in image processing applications [168]. This DL algorithm focuses on spatial inputs and possesses learnable masses and biases, similar to other neural networks. However, since CNNs are primarily designed to process grid topology data, this characteristic distinctly shapes their architecture [159]. Typically, a CNN consist of three main components: the input layer, the convolutional layer, and the pooling layer. The input layer is the most critical element of the CNN. The convolutional layer represents the core of the network, where a multitude of computations are performed. Information flows in a single direction, from inputs to outputs, making the CNN a feedforward network [169]. Each convolutional layer consists of filters that perform convolution operations on the input data, thereby producing feature maps. Deeper networks, with more layers and filters, generally achieve higher accuracy levels [170]. The pooling layer serves to reduce the size of the feature maps. While various pooling techniques exist, max pooling is the most commonly used. This technique works by taking the maximum value from each element of the pooling window [171]. By doing so, it reduces the number of parameters to prevent overfitting and diminish unwanted noise. Additionally, the loss function is employed to measure how far the predicted values deviate from the actual values. A low loss value indicates an increase in the model's accuracy. The

architecture of CNN can be visualised in many configurations. For instance, the architecture developed by the Visual Geometry Group is a well-known model that performs effectively [172]. This architecture includes several feature learning blocks containing multiple consolidated layers, followed by a pooling layer. Recently, various techniques aimed at enhancing CNN performance have been under investigation. Alves et al. [173] examined the impact of data utilisation techniques for improving anomaly classification in PV modules.

Wavelet neural networks

Wavelet neural networks (WNN) stand out from ANN by utilising wavelet transformation as their activation function [174]. WNN comprise three fundamental layers: the input layer, the hidden layer, and the output layer [175]. In the input layer, explanatory variables are transmitted to the WNN, while the hidden layer contains neurons known as "wavelons". These neurons transform the input data into scaled and shifted versions of wavelet functions. The output layer then predicts the target value based on the information received from the hidden layer [176]. WNN's ability to perform multiscale analysis and features like scale shifting make it robust in terms of accuracy and resolution. The wavelet function allows the network to analyse complex and nonlinear systems more effectively, making it a suitable tool for function approximations. The wavelet transformation particularly enhances the WNN's capability for local information extraction, offering significant advantages over traditional activation functions. WNN is built on a theoretical framework based on wavelet decomposition within FNN, serving as a general derivative of RBFN. The core idea of WNN is to employ wavelet functions as the activation function while processing training data, which fundamentally differentiates it from neural networks using classical activation functions [177]. Integrating the wavelet activation function into the hidden layer, combined with multiscale analysis and scale displacement, enables the network to provide higher resolution and precision. One of the primary advantages of this structure is its ability to perform local information analysis on signals and approximate functions effectively. WNN is particularly effective in univariate function approximations. It typically offers a small-scale and computationally efficient structure, facilitating rapid model training. However, due to the limitations of the wavelet activation function, it has historically struggled with complex learning tasks. This issue can be addressed by restructuring the network architecture, specifically by adding convolutional layers, leading to the development of convolutional wavelet neural networks [178].

Deep belief neural networks

Deep belief networks (DBNs) are a type of ANN that functions as a probabilistic generative model, introduced by Geoffrey Hinton in 2006. Designed to handle complex data representations using USL techniques, DBNs consist of stacked sub-blocks based on restricted Boltzmann machines (RBMs), each comprising a visible and a hidden layer [179]. This structure facilitates hierarchical learning, where the hidden layer of each RBM serves as the visible layer for the next RBM [180]. Additionally, DBNs are developed to overcome limitations such as slow convergence, suboptimal learning rates, and local minima commonly found in traditional neural networks [181]. Through this architecture, DBNs effectively learn intricate patterns in data, making them powerful tools for various ML tasks [182].

Radial basis function neural network

RBFN are FNN that consist of three layers and are effectively used for complex data processing and function mapping [183]. The training process occurs in two stages. In the first stage, the masses of the first layer are determined through a USL process using only the input data. The second stage involves the use of both input and output data to establish the masses of the second layer, a process known as SL [184]. One significant advantage of RBFNs is their simpler architecture compared to MLP [185]. This simplicity leads to significantly reduced training times and facilitates the attainment of more reliable results with lower extrapolation errors. Furthermore, RBFNs have a wide range of applications; for instance, they have been successfully utilised in some studies for fault detection in PV systems [186]. Thanks to their straightforward structure and effective learning processes, RBFNs have carved out an important niche in modern AI applications. Consequently, these networks present substantial potential for processing complex data and various applications, as they minimise training times while delivering dependable results. Future research may focus on the integration of these networks with more complex architectures and the development of optimisation methods.

Machine learning

ML is a subfield of AI that focuses on developing systems that learn and evolve from experiences gained through data [187]. In this field, algorithms build models based on example data known as training data and make predictions [188]. By detecting statistical structures among a large number of relevant examples, these algorithms develop rules for automating specific tasks. ML models offer significant potential for making predictions in renewable energy generation,

as they can effectively address highly nonlinear problems [189]. Their ability to adapt to changing environments enables them to outperform traditional statistical models when dealing with new data [190]. The learning algorithms used in ML are typically classified into SL, USL, and semi-supervised learning [191]. To tackle ML problems, data are generally divided into two sets: training data and test data [192]. Initially, a model is coded and trained to analyse relationships within the training data. Once these relationships are established, the model should be capable of solving problems using the test data. The accuracy of the model is assessed by comparing its predictions with the actual outcomes obtained from the test data. If the accuracy is insufficient, methods such as feature tuning or data scaling can be implemented to enhance performance.

Decision tree

DTs are statistical models used for solving both classification and regression problems. These models operate by partitioning data into subsets based on specific rules, allowing for both explanatory and predictive analyses [193]. The fundamental principle focuses on identifying the features that best classify the data, operating within user-defined rules. Consequently, DTs are models that are easy to understand and interpret [193]. They are particularly useful in scenarios where transparency and clarity are crucial, such as in error detection within PV systems [193]. Defined by Breiman [194], this model is characterised by a hierarchical structure composed of nodes and edges. Test functions applied to the data are located at the node points, while the leaf nodes store the final outcomes. During this process, the optimal permutations are identified by minimising the cost function, enhancing the model's effectiveness [195]. The decision tree algorithm uses the "divide and conquer" principle, sequentially breaking down datasets into smaller components. This structural decomposition facilitates the hierarchical construction of the model, branching out from the root node with decision nodes [196]. The "if–then" rules employed in classifying the data come into play throughout the process, and the data processing and learning cycle continues until a final decision is reached [196]. Thus, the model predicts class or value by evaluating specific feature vectors at each node [197]. The criteria for error prediction in DTs encompass factors like split selection and the choice of the optimal tree. These algorithms can work with both continuous and discrete data, providing a broad range of applications [198]. They play a significant role in renewable energy applications, especially in predicting PV solar energy outputs [198]. Additionally, regression-based DTs do not require a predefinition of the model and can adapt to the dataset accordingly [198]. However, as the model becomes more complex, issues such as overfitting and getting stuck in local minima may arise. These problems can be minimised through hyperparameter tuning [198]. In summary, DTs stand out as an effective ML technique for both regression and classification tasks [199]. These algorithms make predictions by branching based on the characteristics of the data until they reach the leaf nodes [199]. In regression tasks, continuous values and time series data are processed, while in classification tasks, the characteristics of events are taken as input [199].

Support vector machine

SVMs, designed to solve binary classification problems, are one of the most effective examples of SL algorithms. This algorithm offers users a model that is both controllable and predictable. Developed by Vapnik in 1995, SVM is based on the principle of structural risk minimisation, aiming to reduce overall risk by minimising the maximum bound of expected risk. Moreover, due to SVM's capability to reduce errors in training data, it can be effectively employed not only for classification tasks but also for regression problems, known as support vector regression (SVR) [200, 201]. Furthermore, SVM stands out as an efficient algorithm for solving various problems, including regression analysis, classification, and the prediction of nonlinear functions [202]. For instance, Kaffash and Deconinck [203] have compared SVM and ANN for hourly day-ahead forecasting, observing that SVM often yielded superior results across most time periods. In addition, SVM excels in detecting visual defects in PV systems by using features extracted from electroluminescence images of PV cells, classifying these faulty images with a high degree of accuracy [193]. SVM can create multiple decisions to prevent overfitting in training data, demonstrating effective performance even with a limited amount of historical data in solar panel disaggregation processes [204, 205]. The primary goal of SVM is to establish a function that exhibits minimal deviation from the true vectors in the training data. The aim is to produce a function that is as smooth and generalisable as possible. SVM not only classifies the data but also possesses the capability to predict which class a sample belongs to with high accuracy. Although various hyperplanes can be created between classes, SVM opts for the optimal hyperplane that maximises the distance between the nearest class members [206]. For complex and nonlinear problems, SVM has emerged as a strong alternative, offering superior accuracy in high-dimensional datasets [207]. Additionally, to manage linearly inseparable data, SVM utilises penalty parameters and kernel functions to successfully map this data into a higher-dimensional space. Thus, the predictive performance of SVM is heavily reliant on the appropriate selection of the kernel function [208, 209]. In conclusion, with its simple structure, minimal parameter requirements, and strong generalisation capacity, SVM

emerges as a favourable choice compared to other statistical algorithms [204]. Its ability to minimise prediction errors and accurately classify data has secured a significant role for SVM in contemporary ML tasks.

Anomaly detection

PV cell irregularities can be defined as sudden deviations in voltage, current, or overall performance resulting from environmental effects [210]. These irregularities may arise due to various reasons, such as the formation of hot spots in individual solar modules, errors stemming from production, shading, or the activation of bypass diodes. At the string level, they may manifest as issues like open circuits, short circuits, or high resistance [211, 212]. Particularly, external factors like dust layers can reduce the amount of light reaching the solar cells, leading to a drop in output voltage and, consequently, losses in the energy produced by the system [213]. Detecting such anomalies is crucial for the long-term reliable operation of solar PV systems and for preventing large-scale failures. Continuous monitoring of systems not only enhances efficiency but also improves the return on investment. Recently, the utilisation of ML techniques for AD has highlighted the significant potential in this area. For instance, one study employed the YOLOv3 model for defect detection on PV panels using infrared and visible light spectrum images captured by drones; this method yielded promising results for automatic fault detection in large solar farms [214]. In another study, a novel DL architecture was developed, combining thermal imaging with edge information, which successfully facilitated the accurate separation of hot spots on PV panels [215]. Additionally, the DL system solAIr has achieved successful outcomes in automatic fault detection in PV farms using thermal imaging [216]. The supervised contrastive learning method demonstrates a high success rate for AD in infrared images of PV modules, requiring no large datasets [217]. Furthermore, a study referenced in [218] emphasised the use of ANN for detecting partial shading situations in PV modules, showcasing how ML techniques can be adapted for such AD. The essence of AD lies in the accurate differentiation between normal fluctuations and genuine faults. Sudden drops in PV current under a constant voltage can serve as a critical indicator for AD, as these voltage variations do not exceed degradation levels. While AD can be performed with a single instantaneous current measurement, achieving a stable voltage level necessitates many more data points [219]. This approach enhances the efficiency of monitoring PV systems and facilitates the early detection of potential faults. Overall, the significant thermal time constant of PV modules results in relatively slow changes in cell temperature, thereby supporting the assumption that the temperature difference between two measurement points is negligibly small [220].

Reinforcement learning

RL is an approach that focuses on complex and uncertain learning processes; in this framework, the model learns from past experiences and feedback to refine its methods and ultimately aims to produce optimal solutions. The primary goal is to maximise rewards in dynamic and often unstable environments, which brings challenges such as balancing conflicting objectives encountered during system design [221]. This learning process can be modelled within a framework known as Markov decision processes (MDP), which defines decision-making mechanisms in variable environments [222, 223]. Within an MDP environment, agents observe the state of the environment, take actions accordingly, and receive rewards or penalties based on their decisions [224]. Over time, through these interactions, the agent learns to develop an optimal policy that maximises long-term rewards [225]. RL stands out as a promising method for optimising complex processes, such as those in energy systems. This technique aims to maximise performance indicators, energy efficiency, cost savings, and reductions in carbon emissions by employing optimal control policies that address the inherent uncertainties and complexities within energy systems [226]. The core of RL can be divided into two main categories: model-based and model-free methods. In model-based RL, the agent uses a model of the environment to simulate the effects of various actions and selects those that will maximise the long-term reward function. This approach is similar to model predictive control (MPC), which also uses system models to predict outcomes and determine control scenarios. However, while model-based RL focuses on maximising long-term rewards, MPC seeks to minimise costs over a specified time frame [227]. Conversely, model-free RL methods operate independently of system models; the agent learns through trial and error via iterative interactions with the environment. In this case, the RL agent adjusts its control policy based on feedback from the environment and gradually learns to select actions that will maximise cumulative rewards over time [228].

Q-learning

RL focuses on complex and uncertain learning processes, where the model learns and improves its methods based on past experiences and feedback, ultimately striving to produce optimal solutions. The primary goal is to maximise rewards in dynamic and sometimes unstable environments, which presents challenges such as balancing conflicting objectives encountered during system design [221]. This learning process can be framed using MDP, which defines decision-making mechanisms in variable environments [222, 223]. Within the MDP framework, agents observe the state of the environment, take actions accordingly, and receive

rewards or penalties based on their decisions [224]. Over time, these interactions allow the agent to learn and develop an optimal policy that maximises long-term rewards [225]. In complex processes such as the optimisation of energy systems, RL emerges as a promising method. This technique aims to maximise performance indicators, energy efficiency, cost savings, and reductions in carbon emissions by utilising optimal control policies developed to address the inherent uncertainty and complexity of energy systems [226]. RL primarily consists of two main categories: model-based and model-free methods. In model-based RL, the agent uses a model of the environment to simulate the effects of various actions and selects actions that maximise the long-term reward function. This approach is similar to MPC, which also employs system models to predict outcomes and determine control scenarios. However, while model-based RL focuses on maximising long-term rewards, MPC aims to minimise costs over a specific time frame [227]. Conversely, model-free RL methods operate independently of a system model; the agent learns through trial and error via iterative interactions with the environment. In this scenario, the RL agent adjusts its control policy based on feedback from the environment and learns to select actions that maximise cumulative rewards over time [228]. Q-learning is one of the fundamental techniques in RL paradigms, recognised as a model-free method that enables an agent to learn a specific task by interacting with its environment [229]. In this process, the agent observes the current state of the environment and performs an appropriate action. After taking the action, the agent receives a reward, and the state of the environment changes accordingly [230]. The ultimate aim of the agent is to develop a decision-making policy that maximises total rewards in the long run [231]. To achieve this goal, Q-learning employs a value function known as the "Q function" to determine the most suitable action for each state. The learning process is continually updated in light of the current state and future rewards through the Bellman equation [232]. However, while traditional Q-learning proves effective in small state and action spaces, its scalability diminishes as the size of the state-action pairs increases [233]. In large-scale problems, the performance of the algorithm may decline, and the learning process may slow down with the expansion of the state space. To overcome such challenges, methods like Deep Q-Networks (DQN), which combine DL techniques, can be employed to enhance the model's effectiveness [234]. In a study conducted by Muriithi and Chowdhury [235], the Q-learning algorithm is applied to energy management in grid-connected solar PV–battery microgrids. Q-learning effectively manages the energy flows of the microgrid, addressing fluctuations in PV production, load demand, and uncertainties in grid tariffs. Through this method, the microgrid management system optimises the interaction between PV production and the ES system, managing the battery's charging and discharging processes and regulating energy exchanges with the grid. Additionally, the Q-learning-based algorithm optimises battery usage, reduces energy costs, and increases the amount of energy sold to the grid. To overcome the limitations of traditional Q-learning, the DQN algorithm, which provides more effective learning in large state-action spaces, is utilised. In the study, the DQN algorithm is tested on loads with different fluctuation profiles in energy management of the PV–battery microgrid, yielding successful results. These results demonstrate that battery charge levels are optimised, grid costs are reduced, and revenues from energy exports are increased.

Genetic algorithm

GAs represent an optimisation method developed by John Holland in 1970, inspired by biological evolution processes [236]. This method is designed based on the mechanism by which organisms adapt to environmental challenges and survive. This mechanism relies on the selection of only the fittest individuals for reproduction, leading to the gradual development of populations over time. The operation of GA consists of three main phases: selection, crossover, and mutation [237]. In this algorithm, each individual symbolises a possible solution to a specific problem, and throughout the process, these individuals evolve. The algorithm, which operates on the principles of natural selection, ensures the survival of the fittest individuals within the population, thereby reaching optimal solutions. The algorithm works with a fixed population, where individuals are evaluated using fitness functions in each generation (reproductive cycle) [238]. The fitness function serves as a tool for measuring the capacity of individuals to solve the designated problem [239]. Individuals with high fitness values are included in the reproductive process to create new generations. Some of the newly produced individuals undergo mutation, and this cycle continues for many generations. Ultimately, only the individuals with the strongest genetic structures persist [240]. GAs can be applied to single-objective or multi-objective optimisation (MOO) problems. Single-objective optimisation concerns itself solely with optimising one objective function, typically resulting in a single solution that aims for either a maximum or minimum value. In contrast, MOO involves the simultaneous optimisation of multiple objective functions, yielding non-dominated solutions. This means achieving a generally better subset of solutions [241, 242]. The first step in applying GAs is determining the population size [144]. The population size can either be randomly selected or optimised based on existing data. In each population size, the value of the objective function is monitored, and the population continues to evolve until better solutions are obtained throughout the process.

Fuzzy logic

FL is employed in two distinct senses. In a narrow sense, it is viewed as a logic system that extends multi-valued logic [243]. In a broader sense, FL is nearly synonymous with fuzzy set theory, which deals with classes of objects that are not defined by precise boundaries and where membership is a matter of degree [244]. From this perspective, FL is considered a subfield of fuzzy set theory. Even the narrow interpretation of FL differs from traditional multi-valued logic systems in both content and structure. One of the fundamental advantages of FL is that it provides an intuitive and flexible solution without creating complexity. This flexibility allows the integration of new functionalities into the system without necessitating changes to the existing structure [245]. FL can model nonlinear and complex functions, accommodating a wide range of input–output data. This process is further facilitated by tools such as adaptive neuro-fuzzy inference systems (ANFIS) [246]. Unlike neural networks, FL systems can utilise the knowledge and experience of human experts. While neural networks often produce opaque models based on training data, FL promotes a more transparent and comprehensible system [247]. Additionally, FL can be integrated with traditional control techniques, enabling more effective solutions without complicating the system. The human language-based nature of FL is also a significant advantage, as it is built upon the linguistic structures used in everyday life, making technology more intuitive and accessible [248]. This technology assesses various parameters by utilising data ranges, which aids in obtaining accurate results. FL comprises two main modules: Mamdani and Sugeno. The Mamdani model divides both input and output parameters into ranges, while in the Sugeno model, input parameters are divided into ranges and output parameters are specified as precise data points [249]. Each parameter's range is expressed through membership functions represented by shapes such as triangles, trapezoids, sine waves, or Gaussians [250]. The model of data variation within this range determines which shape to use, and an appropriate "membership function" is selected for each range [251]. Finally, the fuzzy model attempts to generalise by determining optimal rules based on training and validation datasets. The results obtained are calculated as outputs corresponding to input values using various defuzzification techniques [252]. This method plays a crucial role in predicting accurate results.

Performance optimisation by AI in solar thermal and photovoltaic systems

AI holds significant potential in optimising the performance and efficiency of STSs that harness sunlight to produce thermal energy for various applications. Armghan et al. [253] propose a solar DL model that measures various parameters and solar radiation. Subsequently, an ML algorithm is trained to predict the amount of thermal energy generated. The integration of AI with Internet of Things (IoT) technologies can enhance the performance and energy efficiency of solar cooling panel systems. Results indicate that the synthesised nanoparticles have the potential to improve the efficiency of solar systems due to their unique properties. AI is applied in the context of STSs for optimisation, predictive maintenance, dynamic control strategies, ES optimisation, fault detection and diagnosis, as well as performance prediction and optimisation. Milidonis et al. [254] conduct a comprehensive review of various AI techniques and technologies used in the energy sector, including tasks such as energy storage, monitoring energy efficiency, production forecasting, and overall system design.

Studies on the applications of AI to enhance the efficiency of solar energy systems present various methodologies and results. Cho et al. [255] develop wearable temperature sensors that can predict thermal comfort for homeowners with an accuracy rate of 93.9%. This study demonstrates that the integration of AI applications can improve people's thermal comfort, offering a potential solution to energy consumption issues in buildings. Zhao et al. [256] create a fuzzy RBF neural network through a case study in an office building located in Tianjin to enable the prediction of heating and cooling load estimates. This research highlights the success of AI in achieving high-accuracy predictions that ensure energy efficiency. Xiao et al. [257] employ complex modelling and optimisation techniques to enhance computational efficiency in energy systems, showcasing AI's potential in renewable energy systems. Nwakanma [258] evaluates AI applications in smart homes, investigating the effectiveness of datasets and algorithms, paving the way for the development of new methods for recognising human activities. Olabi et al. [259] address the use of various AI techniques in modelling, optimising, predicting, and controlling the performance of ES systems, particularly thermal ES systems. Commonly used AI approaches include ANN, GAs, particle swarm optimisation (PSO), FL, and ANFIS. The reviews indicate that ANN stands out for its flexibility, low cost, and reliable data processing, while FL-based control systems have proven successful in enhancing energy efficiency. Furthermore, studies employing optimisation techniques like GA and PSO have reported significant gains in cost and efficiency for thermal ES systems. The integration of AI

techniques into ES systems has resulted in notable improvements in overall system performance. Ngarambe et al. [260] highlight the potential to optimise energy usage through AI-based thermal comfort prediction models in building control systems. Zhou et al. [261] provide an in-depth examination of AI applications in heating, ventilation, and air conditioning (HVAC) systems, offering a comprehensive view of control and fault detection. Wang et al. [262] succeed in enhancing energy efficiency in HVAC systems using deep RL. Mehrjardi et al. [263] utilise generative adversarial networks (GANs) and limited data obtained from numerical simulations to optimise fin configurations in a square structure containing PCM within ES systems. The study combined GANs with CFD data to create reliable new datasets, determining an optimal fin spacing of 15.75 mm and an angle of − 11.6°, significantly reducing computational costs and shortening the melting time of PCM by approximately 50%. The integration of GANs and CFD has led to substantial advancements in evaluating the performance of thermal systems, yielding results highly applicable in industry and engineering. Shaban et al. [264] explore the potential for enhancing efficiency in photovoltaic-thermal (PV/T) systems using deep neural networks (DNN). Boretti [265] demonstrates that STSs provide a more environmentally friendly alternative compared to battery storage. Moustafa et al. [266] aim to improve the performance of a tubular solar distillation unit equipped with an electric heater, resulting in a 31.85% increase in water yield and an energy efficiency of 38.61%. Additionally, an ANN model optimised through meta-heuristic optimisation (humpback whale optimiser) achieved higher prediction accuracy compared to traditional models. Overall, this study underscores the significance of AI and meta-heuristic methods in improving the thermal performance of solar distillation systems. Ghritlahre and Prasad [267] employ an FNN model to predict air conduction through rough absorbent plates in an SAH, comparing it to experimental data. The backpropagation MLP model is developed using five input parameters (number of rough surfaces, relative roughness height, roughness spacing, roughness size, and Reynolds number) and one output parameter (Nusselt number). The ten-neuron model was found optimal based on statistical error analysis and yielded successful results in heat transfer predictions, outperforming the multiple linear regression (MLR) model by 1.79%. Al-Waeli et al. [268] present a study assessing the performance of PV/T systems, focussing on testing different PV/T configurations, traditional PV, water-based PVT, water-nanofluid PV/T, and nanofluid/nano-PCM, using an ANN-based MLP under the same conditions. The results indicate that employing nanofluid/nano-PCM increases electrical efficiency from 8.07 to 13.32% and achieves a thermal efficiency of 72%. This study emphasises the use of ANN models in predicting the performance of PV/T systems and managing future solar energy applications. Rehman et al. [269] conduct a study using DNN and explainable AI methods to predict the performance of SAH. The research examined a double-pass SAH system equipped with circular and semicircular edge-finned aluminium tubes filled with RT44HC and RT18HC energy storage materials. Parameter data collected from three different heater configurations enabled high-accuracy heat output predictions ($R^2 = 0.995$, RMSE = 0.018). The developed DNN model, combined with GAs, identified optimal parameter combinations that achieved a 76.328% efficiency. The findings revealed that temperature increase was the most influential factor on heat output, followed by absorber plate temperature, heat input, exposure time, ambient temperature, and solar radiation. These results have significant implications for the design, operation, and maintenance of SAH systems. Assareh et al. [270] conduct a study on the analysis and optimisation of a solar energy system integrating cooling, hot water, and electricity generation. The research employed response surface methodology (RSM) to achieve MOO using PV/T modules for generating electrical and thermal energy. The findings indicated that the optimal axis efficiency of the system was 19.704% with an appropriate cost ratio of $1.774/h. Moreover, evaluating the system across six climatic regions in Australia identified Maitland City as the most suitable location for system application. This study underscores the importance of climate-based analysis in the design and implementation of solar energy systems, contributing to sustainable energy solutions. Tilahun's research [271] examines the integration of STSs in process industries, noting that while they provide economic benefits, they also introduce uncertainties and fluctuations in energy systems. The study developed a framework for optimising complex and dynamic STS design and management strategies using ML methods. This approach reduces oversizing by 10–30% while enhancing energy efficiency in the solar field and storage by 23.7% and 8.3%, respectively. The study emphasises the potential of ML for industrial STSs while offering various recommendations for future research.

Al Smadi et al. [272] emphasise the growing importance of PV systems in light of the necessity to address climate change and reduce carbon emissions. Their study discusses the significance of employing AI to overcome the inadequacies of energy storage batteries and proportional-integral controllers that regulate active and reactive power, ultimately optimising energy production in PV systems. As a novel approach, the integration of ANFIS with solar PV systems is explored, highlighting the role of these systems in energy management. The findings present promising results in terms of energy regulation and management, underscoring the critical importance of AI in enhancing the performance of PV systems. Similarly, Ganesan et al. [273] stress that detecting faults in PV arrays is crucial for enhancing the safety and reliability of PV systems. Their study aims

to achieve more accurate fault detection using AI-based techniques, including fuzzy inference, wavelet, SVMs, and k-nearest neighbours. Unlike traditional methods, the proposed approach can identify faults with minimal sensor usage. Experimental results indicate that fault detection has led to a power gain of up to 152%, with the power output of the PV array increasing from 62.65 to 160 W. Moreover, Sarang et al. [274] analyse the efficiency, cost, complexity, stability, and shading of maximum power point tracking (MPPT) techniques in PV systems. Their study reveals that traditional MPPT techniques achieve an accuracy between 94.3 and 99.6%, while AI-based techniques attain accuracy rates ranging from 98 to 98.8%. Among AI-based controllers, ANFIS and ANN demonstrate the highest performance, whereas traditional PSO and incremental conductance PSO techniques also yield robust results. This research aims to identify the optimal MPPT strategies that will enhance energy efficiency in PV systems. The study's findings suggest that AI-based ANFIS and ANN techniques exhibit superior performance, with FL control emerging as the most cost-effective option regarding efficiency. Additionally, Marlin and Jebaseelan [275] investigate the effects of meta-heuristic optimisation algorithms on MPPT in PV systems. They compare three different optimisation techniques: mongoose optimisation (MO), prairie dog optimisation algorithm (PDOA), and a hybrid PDOA + MO method. The authors note that traditional MPPT methods often fail to effectively track the global maximum power point (GMPP) under varying environmental conditions, while the AI-based approaches developed in this study yield better results. Notably, the hybrid PDOA + MO model achieves an impressive efficiency of 99.5%, with a rapid rise time of just 1.6 s and a low error rate of 0.48, demonstrating superior performance compared to the other methods. Águila-León et al. [276] explore the effectiveness of a grey wolf optimiser (GWO) optimised PSO algorithm for MPPT control in PV systems. Their research indicates that this algorithm can increase energy production by up to 20% compared to traditional MPPT algorithms. Simulations reveal that the GWO-optimised PSO exhibits high efficiency and stability under varying solar irradiance conditions, highlighting the potential of this algorithm in energy engineering applications. Furthermore, Mohanty et al. [277] investigate the optimisation of the Nusselt number and thermal efficiency for a three-sided embossed SAH. The study demonstrates that the best results obtained using the multi-objective particle swarm optimisation (MOPSO) algorithm and RSM yield Nusselt numbers between 65 and 80 and thermal efficiencies ranging from 63 to 75%. The optimisation results, consistent with experimental data, have been validated using maximum deviation theory, with a margin of error of just 3.39%. This work aims to enhance the effectiveness of SAHs for engineers, stressing the need to consider various responses in future research. Wu et al. [278] focus on the geometric analysis of parabolic trough SCs of varying sizes. They propose a Python-based algorithm to calculate optical efficiency based on a secondary hyperbolic mirror. The results indicate that the error rate between the simulated and theoretical optical efficiency is less than 3.13%, while the ML-supported model achieves a fit of 97.874 R^2 for optical efficiency. This study offers a significant computational method for the geometric design and energy analysis of PTCs, demonstrating the proposed model's superiority over existing models. He et al. [279] address the challenge of balancing economic benefits and environmental protection in the design of solar heating systems, emphasising the importance of MOO approaches. Their study combines the MOO of SC area and thermal ES tank volume using TRNSYS simulation, ML, and GAs. The research demonstrates that CO_2 emission reduction is maximised by increasing SC area and TES tank volume, whereas single-objective optimisation fails to reveal these interactions. Consequently, MOO provides better options for solar heating system designers, allowing for a more effective balance of environmental and economic benefits. Maharana et al. [280] propose a model to determine the optimal performance and design conditions for FPSWHs. This model, which predicts thermal, optical, and exergy efficiency using hourly solar radiation data, is supported by various computational intelligence techniques. Among the methods used for single-objective optimisation, the water cycle algorithm shows superior performance across all statistical parameters. Additionally, MOO is employed to achieve a balance between maximum exergy efficiency and minimum absorber area. Ultimately, a MATLAB-based GUI tool has been developed for design optimisation, indicating the potential of this study to optimise the performance of solar combi systems. Vaziri et al. [281] experimentally investigate the performance of cost-effective double-pass perforated glass SAHs under various operating conditions. Their research shows that thermal efficiency increases with rising airflow, with the highest efficiency reaching 94% for DPGSAHM and 97% for DPGSAHI. Notably, DPGSAHI exhibits higher performance than DPGSAHM at the same airflow rate (0.024 kg s^{-1}), achieving 89.3% efficiency compared to approximately 80% for DPGSAHM. When employing a hybrid optimisation method (Taguchi-ANN), thermal efficiency increases by 1.25%, and exergy efficiency rises by 2.4%. The results reveal that the energy cost (0.028 $/kW) for both types of heaters is higher than that of other double-pass heaters in the literature. In the work of Liu et al. [282], the energy demand of a 220 m^2 zero-energy building is optimised using nanocomposite solar panels. The solar panel modelling achieves a high-accuracy rate of 99.98% through hybrid ML methods. The properties of nanomaterial solar cells are predicted using SVMs and the k-nearest neighbour algorithm (KNN) across four different seasons. The KNN algorithm is further

optimised with PSO, resulting in a remarkable 170% increase in solar cell efficiency. This study demonstrates the potential of nanocomposite materials and optimised cell structures to significantly enhance solar panel performance in commercial and residential buildings. Chouicha et al. [283] aim to maximise energy production in PV systems. They develop a new MPPT method using an FL control algorithm to enhance the efficiency of solar panels. The FLC-based MPPT controller, simulated in the MATLAB/SIMULINK environment, effectively tracks maximum power under varying solar irradiance and temperature conditions. The research shows the potential of FL control to optimise power output in PV systems. Yang et al. [284] seek to optimise research aimed at producing high-performance solar cells in light of global trends in solar energy. They propose the chimp optimisation algorithm (ChOA) for modelling and determining solar cell parameters. ChOA stands out with a faster convergence rate, global search capacity, and robustness compared to existing optimisation algorithms. Utilising this algorithm to determine the parameters of different PV modules and solar cells has demonstrated high accuracy and performance through extensive comparative studies. Moreover, ChOA has been successfully applied to commercial modules (KC200GT, SW255, SM55), showing robust results against changes in temperature and irradiance. Kumar et al. [285] have developed a hybrid algorithm, PSO_ML-FSSO, for MPPT in solar PV systems. This algorithm combines PSO with ML trained techniques and flying squirrel search optimisation (FSSO) to achieve high efficiency under varying conditions such as changing temperature, solar irradiance, and partial shading. Comparative simulations conducted in MATLAB/Simulink reveal that this new algorithm enhances efficiency by 0.72% compared to other popular MPPT methods while reducing settling time by 76.4%. De and De's study [286] examines the use of AI and ML techniques for the integration of solar PV systems with the grid. The research emphasises that traditional methods are being replaced by big data technologies and ML algorithms, which yield more efficient results. The investigations indicate that employing ML facilitates more effective decision-making regarding significant factors such as load demand forecasting and the optimal sizing and placement of solar PV systems. It is underscored that these techniques possess considerable potential to improvethe integration of solar PV systems into the grid and enhance energy efficiency. Alshareef's work [287] focuses on the performance degradation caused by partial shading conditions (PSC) in large-scale solar PV systems. This study proposes a new strategy based on the falcon optimisation algorithm (FOA) for GMPPT as a solution to the multiple peak power points (P–V curves) that arise under partial shading. The FOA addresses the shortcomings of commonly used algorithms such as perturb and observe (P&O) and PSO, including slow convergence rates and oscillations in steady-state conditions. According to simulation results, FOA achieves over 99% efficiency in MPPT and demonstrates superior performance with rapid tracking times compared to other algorithms. This study reveals that FOA is an effective tool for performance optimisation in solar PV systems. Azli et al. [288] propose a novel MPPT algorithm called the yellow saddlefish algorithm (YSGA) to optimise the efficiency of solar PV systems under PSC. This method exhibits higher speed, accuracy, and robustness compared to other known algorithms. Simulations and statistical analyses indicate that the YSGA demonstrates superior performance in power tracking within PV systems. The study by Khan and Pushparaj [289] investigates MPPT methods for PV systems. It presents a comparative analysis of P&O, FL, and a proposed hybrid MPPT algorithm, focussing on performance optimisation under varying irradiance conditions. Simulation results reveal that the proposed hybrid method tracks the maximum power point more accurately and reduces fluctuations. The hybrid MPPT method is presented as a viable option for increasing energy production efficiency, especially in rural areas. Roy et al. [290] utilise three different ANN-based algorithms, Levenberg–Marquardt, Bayesian regularisation, and scaled conjugate gradient (SCG), to obtain MPPT energy in solar PV systems. The performance of these algorithms is analysed comparatively to determine the most suitable algorithm. Simulations are conducted with the ANN trained on a dataset of 1000 samples, and results indicate that the SCG algorithm generally performs better, while the LM algorithm excels in processing the dataset. In the study by Ibrahim et al. [291], a mathematical model is restructured and validated to optimise the performance of PV systems under PSC. The study improves the dynamic response and steady-state performance of PV systems using heuristic algorithms such as the cuckoo search algorithm (CSA) and modified PSO. Simulation results demonstrate that the CSA effectively distinguishes global and local MPPs, operating with 99% efficiency. This approach is proposed to enhance the performance of PV systems, particularly under PSCs. Balal et al. [292] develop a new method by combining two different approaches to address the drop in power production and the multiple peak problems in P–V characteristics caused by PSC. Firstly, a distributed MPPT method is employed to independently manage each PV module or module group, facilitating efficient solar energy harvesting. Secondly, an optimised fuzzy/bees algorithm is utilised to minimise power fluctuations arising from conventional methods without relying on temperature and light sensors. Simulations show that this approach enhances the efficiency of PV systems, providing approximately a 4% increase in power in large solar farms. Gopi et al. [293] present AI-based forecasting models for a 2 MWp solar PV plant in Kerala, India, using irradiance, wind speed, and temperature as inputs.

Among the tested approaches, ANFIS achieved the highest accuracy ($R^2 = 0.9901$, Nash–Sutcliffe efficiency = 0.9828), demonstrating strong potential for predicting annual energy yield and performance ratio, thereby supporting policymakers and developers in solar farm planning. Khenissi et al. [294] develop an optimum grid-connected PV system sizing method for a typical home in Tunisia. The study employs two optimisation methods, PSO and GA, to maximise net present value (NPV) and minimise the loss of power supply probability (LPSP). Simulation results indicate that the PSO algorithm outperforms the GA in terms of speed and optimal solution quality. The findings reveal that the system maximising NPV is more profitable and reliable, consisting of 56 PV modules, 7 batteries, and 2 inverters. El-Sebah's study [295] highlights the increasing importance of solar energy in electricity generation and the need for effective power extraction due to the low conversion efficiency of PV systems. In this context, three different MPPT controllers are implemented: sliding mode control (SMC), FL controller, and ANN. The performance of these controllers is assessed under varying irradiance and temperature conditions. The results demonstrate that SMC outperforms FLC and ANN with lower oscillations. Belhaouas et al. [296] focus on reducing the adverse effects of partial shading on the performance of PV generators. The proposed new PV system structure aims to mitigate the impacts of partial shading by eliminating local MPPs and enhancing power output. The study compares the effectiveness of MPPT techniques with various PV configurations, proposing the most suitable and cost-effective solution. Overall, the study provides significant contributions to the optimisation of PV system performance. Sharma et al. [297] address the optimisation role of MPPT algorithms in solar PV energy production. The study highlights that an ANN-based MPPT system can enhance the performance of MPPT under rapidly changing atmospheric conditions. Results obtained through the use of ANN demonstrate a superior ability to determine the MPP more quickly and accurately compared to other methods. Consequently, the study presents an effective and precise way to extract maximum power from solar panels using a DC-DC boost converter, indicating significant contributions to performance optimisation. Hashim et al. [298] note that PV panels experience peak power loss under clouds or shadows. This study aims to maximise performance from solar energy systems by using a hybrid approach enhanced by ANN, considering the environmental impacts that reduce the efficiency of PV systems due to partial shading. Tests conducted on the MATLAB/Simulink platform compare the efficiency of the proposed hybrid solution with other methods, evaluating the best options regarding current, voltage, and power. The results demonstrate that the suggested ANN approach improves PV system performance under PSC achieving maximum power production exceeding 7 kW. Zafar et al. [299] highlight the issues of power loss and efficiency decline associated with MPPT control algorithms in solar energy systems, particularly under PSC. Among the bio-inspired optimisation algorithms commonly used in the literature are grasshopper optimisation, GWO, PSO, cuckoo search, and PSO-gravitational search. However, these methods face disadvantages, such as high tracking and settling times, as well as oscillation problems during global maximum (GM) tracking. Studies in the literature emphasise the need for the development of new optimisation techniques that can provide more effective and rapid power tracking under PS conditions. The work of Saeed et al. [300] proposes a method for MPPT using a multi-layer feedforward ANN. It has been demonstrated that the ANN-based approach is more effective than traditional P&O and incremental conductance algorithms, particularly in reducing energy losses in situations like shading. The study reveals that the ANN method enhances the efficiency of both PV modules and solar battery charge controllers, showcasing its superiority over conventional MPPT algorithms. Almadhor et al. [301] propose a hybrid algorithm for optimising the capacity configuration of a solar PV–battery-based microgrid. This algorithm aims to optimise the solar energy production site and capacity by combining the advantages of PSO and bat algorithm. The rapid convergence capability of PSO and the low computation time advantage of BA have reduced energy losses in the system, thereby enhancing the efficiency of the microgrid. The results indicate that the hybrid algorithm optimises system performance by minimising active power transmission losses. The work of Padmavathi et al. [302] presents a regression-controlled approach for MPPT in solar PV systems. Due to the poor performance of traditional MPPT algorithms (P&O, FPA, and PSO) under partial shading and rapid isolation changes, a regression-based controller has been developed. This controller predicts the necessary duty cycle for the boost converter by utilising PV system data, making MPPT more efficient. Simulation and hardware tests show that the regression controller is 15% to 20% more efficient than conventional algorithms. In the study by Nagadurga et al. [303], the cat swarm optimisation (CSO) method is proposed for obtaining the MPP from solar PV systems under PSC. This study aims to address the inadequacies of traditional MPPT techniques in locating global peaks, highlighting the optimisation power of CSO. According to simulation results, the CSO algorithm provides a faster tracking speed and higher accuracy compared to the PSO algorithm. CSO helps to reduce power losses in PV systems due to its capability to track GMPPs more accurately.Abo-Khalil et al. [304] propose an opposition-based firefly algorithm (OFA) as an alternative to traditional methods for optimising MPPT in PV systems. This algorithm enables PV systems to track the GMPP more quickly and efficiently, both under normal irradiation conditions and

during partial shading. The firefly algorithm has been optimised to enhance tracking speed by utilising its capability to search in the opposite direction. Simulations and experimental tests show that the proposed algorithm yields faster, and more accurate results compared to the traditional P&O method. The OFA algorithm operates with nearly 100% efficiency even under PSC, resulting in less power fluctuation. Ye et al. [305] propose an ANN-assisted method for optimising the scaling factor in variable step size (VSS) MPPT algorithms. This method employs an ANN to determine an appropriate scaling factor, aiming to improve the performance of the traditional VSS incremental conductance (IncCond) MPPT algorithm under rapidly changing irradiation conditions. This approach calculates the optimal scaling factor using data from two operating points without requiring expensive irradiation and temperature sensors. Experimental results demonstrate that this method significantly reduces tracking loss and enhances tracking accuracy compared to other methods under both variable irradiation conditions and EN50530 test conditions. Chao and Rizal's study [306] develop a hybrid MPPT controller. This controller aims to enhance the performance of PV systems by combining GA and ant colony optimisation (ACO) methods. The research demonstrates that, under various shading conditions, the GA-ACO controller reaches the GMPP more quickly and accurately than traditional P&O and ACO methods. Results indicate that the GA-ACO approach operates with fewer iterations and greater accuracy in determining the MPP, representing a significant advancement in improving the overall efficiency of PV systems. Ibnelouad et al. [307] propose an ANN-PSO approach to enhance the efficiency of PV energy systems. This hybrid method uses ANN to predict solar irradiance and cell temperature, while PSO optimises power generation by tracking the MPP in real time. Simulation results conducted in MATLAB/Simulink demonstrate that the ANN-PSO method successfully extracts up to 97% of the maximum power despite significant fluctuations in environmental conditions. The study emphasises the potential of the ANN-PSO approach to improve the performance of PV systems and highlights its rapid tracking speed, which significantly enhances system effectiveness. Al-Majidi's research [308] focuses on MPPT techniques in PV systems. Various methods, including P&O, FL control, ANFIS, and ANN, have been developed and compared for their performance. The proposed methods demonstrate high tracking efficiency, particularly under varying atmospheric conditions, with the FLC and P&O-based MPPT methods effectively tracking the MPP under fluctuating solar irradiance. Consequently, improvements made to the P&O algorithm and the implementation of AI-based techniques have increased the efficiency of PV systems to an impressive 99.6%. Khan's study [309] examines MPPT methods in solar PV systems. The proposed AI-P&O control method enhances the power, voltage, and current performance of the PV system, achieving values of 272.4 W, 157 V, and 1.74 A compared to various MPPT techniques. Simulations indicate that the AIAPO method yields more accurate and stable results than the P&O, FL, and ANN methods. This research highlights the potential of the hybrid control approach developed to track the MPP under variable irradiation conditions, thus improving the efficiency of PV systems. Ram et al. [310] aim to develop reliable and practical MPPT techniques to maximise the power output of PV power plants. The study proposes a new bio-inspired fast pollen algorithm (FPA) combined with the conventional P&O method. The FPA-P&O algorithm demonstrates a rapid convergence rate with less switching stress compared to ELPSO and PSO methods under four different shading conditions, particularly reaching the GMPP in 0.45 s, ensuring high power conversion efficiency. The findings indicate that FPA-P&O serves as an effective alternative for existing hybrid methods and provides a suitable solution for residential rooftop PV installations. Almutairi et al. [311] propose the GWO method as a GMPPT algorithm to overcome shading issues in PV systems. The study shows the implementation of an MPPT controller based on the GWO algorithm, along with its integration to maximise energy output. Experimental tests have been conducted to validate the effectiveness of the proposed OGWO-MPPT method under different solar irradiation conditions and partial shading scenarios. Results demonstrate that the OGWO-MPPT method possesses a shorter convergence time and less power fluctuation compared to the P&O method. Future studies aim to evaluate the performance of this algorithm in various PV grid-connected system topologies. Abdallah et al. [312] highlight the critical role of tilt and azimuth angles in enhancing the efficiency of PV systems amidst rising solar energy demand. In this study, the optimum tilt and azimuth angles for PV panels in all countries are determined using the photovoltaic geographic information system. The developed ANN model accurately predicts annual radiation alongside optimum tilt and azimuth angles. Research findings emphasise the necessity for solar panels in the northern hemisphere to face south, while in the southern hemisphere, they should face north. Additionally, it is indicated that the tilt angle should be equal to or greater than the latitude value for latitudes up to 30°, while it should be less than 30° thereafter. The study concludes that the proposed neural network model can be integrated into low-cost microcontrollers and utilised for PV system analyses. Mansoor et al. [313] address the issue of MPPT in PV systems by proposing a new MPPT controller based on Harris hawk optimisation (HHO). The study reveals that the HHO-based MPPT outperforms traditional methods such as P&O, DFOA, PSO, and GWO in tracking maximum power under shading conditions. Analyses indicate that HHO provides a 10–30% faster tracking time and a reduction in oscillation of

up to 95.45%. These results validate HHO as an effective solution for enhancing the energy efficiency of PV systems. Sohani et al. [314] thoroughly investigate the effects of absolute and relative humidity on the performance of PV solar modules. By employing ANN, models that predict voltage and current are developed, taking into account effective parameters such as temperature, humidity, wind speed, and irradiation. Parametric studies evaluate the impacts of absolute and relative humidity on key characteristics like voltage, current, power, and efficiency. Results indicate a declining trend in all performance criteria with increasing absolute and relative humidity; the polycrystalline module was found to be more sensitive than the monocrystalline module, with power and efficiency variations of up to 46.3% observed within a specific humidity range. The developed ANN models contribute to more accurate predictions of PV module performance, thereby aiding in clean electricity generation. Cipriani et al. [315] propose an innovative approach that combines thermographic non-destructive testing with AI techniques to classify losses in PV systems. Low electricity production is typically attributed to a decrease in efficiency caused by dirt and external factors on module surfaces. The proposed system employs CNN to automatically classify thermographic images with 98% accuracy, facilitating rapid diagnosis, cost reduction, and minimisation of electricity production losses. Ndiaye and Faye [316] present a study that experimentally validates two soft computing techniques for MPPT. The research uses PSO and ANFIS to conduct simulations with data obtained from a PV array installed at Polytechnic High School in Dakar. Results show that the ANFIS method exhibits the best performance, characterised by smaller error margins and shorter response times. Overall, the ANFIS method stands out as a robust and effective approach for optimum power generation. Hamidi et al. [317] discuss a gradient-based optimisation algorithm developed to achieve maximum energy output from PV systems. The study formulates an optimisation problem using the Nesterov accelerated gradient framework for MPPT. Furthermore, the Adadelta approach is adopted to determine the step size. The findings reveal that the proposed method outperforms the commonly recommended P&O method. The research highlights the potential of the proposed method to enhance the performance of PV systems and suggests future investigations into topics such as the effects of shading. Tavares-Cachat et al. [318] focus on the efficient use of solar energy in BIPV for building facades. The research optimises the design of fixed external louvre photovoltaic shading devices using MOO and integrated simulations. The results demonstrate that configurations with fewer louvres enhance thermal, electrical, and lighting performance, thus optimising the use of solar energy. The proposed methodology has the potential to overcome the limitations of standard façade configurations and promote collaboration between researchers and the industry.

Performance prediction by AI in solar thermal and photovoltaic systems

The integration of AI into STSs has emerged as a transformative approach for enhancing performance prediction. By leveraging advanced algorithms and data analytics, AI enables more accurate forecasting of system efficiency and energy output. This development not only optimises operational parameters but also contributes to the overall sustainability of renewable energy solutions.

Dikmen et al. [319] investigate the use of ANN and ANFIS to predict the thermal performance of vacuum tube SCs. The results indicate that the ANN method exhibits superior performance, achieving a satisfactory R^2 value of 0.811914. Furthermore, new mathematical formulations derived from the ANN model are presented to enhance the accuracy of thermal performance predictions. The optimisation of system parameters using GAs demonstrates the effectiveness of these methods in improving the thermal performance of SCs. Similarly, Yaïci and Entchev [320] conduct a detailed examination of the applicability of ANFIS in predicting the performance parameters of Solar TESs. In their study, an ANFIS network is developed and optimised using experimental data collected from various seasons and weather conditions in Canada. The obtained predictions show an excellent agreement with the experimental data, with a relative error of less than 1% for stratification temperatures and 9% for solar ratios. The results highlight the high accuracy, efficiency, and reliability of the ANFIS method, which slightly outperforms the ANN model. Mandal et al. [321] perform a thermophysical analysis of the Manzanares solar chimney plant by optimising collector inlet height, chimney diameter, and divergence under varying irradiation levels, achieving up to 318% power enhancement. A comparative evaluation of machine learning models further reveals that Gaussian process regression consistently provides the highest predictive accuracy, outperforming ANN, SVM, and other techniques. Çerçi et al. [322] examine the performance of SAHs, commonly used for obtaining thermal energy from solar energy systems. In their research, two different SAH models, V-type (with V-corrugation) and B-type (with internal compartment assembly), are designed, produced, and experimentally tested under the climatic conditions of Osmaniye, sharing the same external volume (same base area and height). The experimental results show that the V-type SAH provides higher temperatures, energy efficiency, and exergy efficiency compared to the B-type SAH. Additionally, various ANN models are developed to predict the performance parameters of both SAHs. These

results reveal that the developed ANN models can accurately predict the performance of SAHs, offering valuable insights for researchers. Odufuwa et al. [323] emphasise the importance of effectively predicting thermal system performance for the optimisation of energy systems. Their study develops an ANN model to predict the performance of Ice Thermal ES systems for HVAC systems. The developed model predicts performance parameters such as storage temperature, coefficient of performance, cooling thermal load, and refrigerant power consumption. By employing a comprehensive dataset that includes various meteorological and physical data, the study enhances the accuracy of the ANN model. The results demonstrate that the model can make predictions with high accuracy, with R values reaching up to 0.99 for training, validation, and testing. The findings highlight the application potential of the ANN model in HVAC and ITES systems and illuminate future research directions aimed at improving energy consumption efficiency. Kumar et al. [324] investigate the performance of a jet-impingement solar thermal air collector equipped with multiple curved edges. They employ methods such as linear regression, Gaussian process, and random forest to predict parameters like Nusselt number, friction factor, and thermohydraulic performance, finding that the Gaussian process model outperforms the others with the highest accuracy rate. The results provide significant data to enhance the efficiency of such collectors. In their study, Alawi et al. [325] stress the importance of exergy analysis in solar thermal applications, developing six different ML models to predict the exergy efficiency of parabolic trough collectors. The developed models include adaptive boosting (AdaBoost), multivariate adaptive regression splines, stochastic gradient descent, Tweedie regressor, and ensemble learning techniques like voting and stacking. The results show that the stacking regression model predicts exergy efficiency with the highest R^2 value of 0.963, followed by the AdaBoost algorithm with a value of 0.947. The study emphasises the need for future investigations involving a broader set of input parameters and the comparability of the current model with other models. In a study conducted by Suman et al. [326], an innovative CFD and DNN approach is developed to accurately predict the performance of an artificially roughened SAH. The optimised DNN model, with a 3–37–37–1 architecture, predicts the thermal–hydraulic performance factor, with maximum, average, and minimum error rates of 2.54%, 0.245%, and 0.0009%, respectively. A global sensitivity analysis conducted using Shapley values indicates that the rib height is the most effective design parameter. The generalisation capacity of this model is evaluated by comparing it with experimental data, revealing that the optimised DNN model provides superior prediction performance compared to traditional CFD simulations. Finally, Kocer [327] investigates the effect of artificial roughness (the presence of repeating edges) on the heat transfer and thermal–hydraulic performance of SAHs. Numerical analyses are performed on overheat flow using three types of ribs with different geometries at Reynolds numbers ranging from 4000 to 18,000 and various relative roughness ratios (7.14, 10.71, and 14.29). ML techniques are employed to predict the Nusselt number, friction factor, and thermal–hydraulic performance parameters. The results indicate that the highest Nusselt number is achieved at a relative roughness of 7.14 and a Reynolds number of 7500, with the XGB model providing the most accurate predictions for both the Nusselt number and friction factor. GaneshKumar et al. [328] investigate a SAH system enhanced by implementing V-curvature, surface roughening (shot blasting), nanocoating, and the inclusion of black pebble stones (BPS) to increase the surface area. The surface was roughened through shot blasting and subsequently coated with activated carbon-based nanomaterials. The use of BPS as thermal mass contributed to enhancing the thermal efficiency of the SAH. ML models are employed in the study to assess the impacts of surface modifications and BPS, with the random forest regression (RFR) model yielding the best results at an accuracy of 99%. This method offers promising pathways for improving energy efficiency in SAH systems. In the work by Sridharan and Balaji [329], the thermal efficiency of a solar FP water heater system is predicted using an ANFIS. The thermal power output and incoming solar radiation serve as inputs for predicting thermal efficiency. Experiments are conducted with three different configurations (independent, series-connected, and parallel-connected), maintaining a constant fluid flow rate. The results indicate that the parallel-connected system records the highest thermal efficiency at 26.51%. The three developed ANFIS models are trained with experimental data and successfully predicted the experimental results with an accuracy of 99.89%. Lillo-Bravo et al. [330] proposed an RF model to replace experimental tests for predicting annual energy supply and solar fraction for a thermosiphon solar water heating system. This model, which can substitute for experimental tests, achieved high accuracy with a mean absolute percentage error (MAPE) ranging from 2.94 to 5.86% and an R^2 value exceeding 99%. This approach contributes to reducing the cost and time of experiments, facilitating the commercialisation of renewable energy technologies and providing system designers with an important tool for performance prediction. Moghadasi et al. [331] develop various ML models to predict the output air temperature, humidity, and thermal efficiency of an SAH, suggesting a model for optimal control of the SAH. The study, based on over 62,500 data points obtained from experiments conducted in the winter of 2022, evaluates the performance of different algorithms. The LightGBM algorithm delivered the highest accuracy, achieving an R^2 of 98.27% and a MAPE of 2.95% on the test set. The study demonstrated that the model-based control

strategy for the SAH could yield 24% more thermal energy during a year-long winter season. Saravanan et al. [332] utilise ML models, including RF, LR, and KNN, to predict the thermal performance of a C-shaped finned SAH. Among the models trained with experimental data, the RF model is identified as the most successful, exhibiting the lowest error and highest R^2 value. The RF model predicts thermal performance parameters with 98% accuracy, while the performance of the other two models is comparatively lower. This study highlights the effectiveness of the RF model as a tool for modelling the thermal performance of SAH systems. In the study by Kuang et al. [333], a CNN approach is employed to predict the performance of a flat glass vacuum tube SC. The CNN model outperforms other prediction models, namely MLR and back propagation (BP) neural network models, achieving the lowest root meansquare error (RMSE = 0.00577), the lowest absolute error (MAE = 0.00357), and the highest coefficient of determination ($R^2 = 0.9534$) in predicting collector efficiency. The CNN model indicates that solar radiation and mass flow rate have the most significant impact on collector performance. Badji et al. [334] focus on the use of renewable energy for greenhouse heating, establishing an optimal environment for plant growth through a unique SAH and underground latent heat storage system. The experiments demonstrated that the greenhouse equipped with the heating system maintained a temperature 57% higher overnight compared to a conventional greenhouse. Predictions of internal greenhouse temperatures are made using ANN, with the NARX model yielding high accuracy ($R^2 = 0.9986$). Moayad and Krisztián [335] employ the ANN technique to predict the energy performance of an SAH. The model, trained on data obtained from Miskolc, Hungary, predicted the performance of the V-corrugated SAH using 336 datasets with seven input and one output parameter. The best results are achieved using eight neurons in the hidden layer, demonstrating a high correlation with actual values. This ANN model offers an effective alternative for predicting the thermal performance of SAH systems in terms of time and cost efficiency. Wei and Calautit [336] investigate price-sensitive MPC strategies on an integrated STHS for buildings with a variable workforce structure. The study conducted at a university building in Nottingham, UK, shows that the MPC control strategy optimises solar energy utilisation, reducing energy costs by 50% while ensuring indoor comfort. Additionally, the developed model increases the use of low- and medium-priced energy by 58%, providing greater comfort alongside ES systems and significantly reducing electricity costs. Osorio et al. [337] develop a deep operator network (DeepONet) based ML model to predict the dynamic behaviour of thermal systems operating under variable energy inputs and load requirements. The developed model accurately predicted the state-of-charge (SoC) and efficiency of a solar energy-based system, outperforming the conventional LSTM model. DeepONet emerges as a significant alternative for optimising such systems, offering high-accuracy predictions at low computational costs. Du et al. [338] combine ANN models with dataset filtering to develop a method for predicting the thermal efficiency of SCs. In the research, data are filtered through clustering analysis, and ANN models are trained and tested. The study significantly improves the prediction accuracy of ANN models with complete data filtering. The CNN model provided higher accuracy than the BP model, maintaining the highest prediction accuracy across all weather conditions. Cuce et al. [339] investigate the optimal design of solar chimney power plants through a 3D CFD model validated against experimental data from the Manzanares pilot plant. The study identifies 0.6° collector slope and 1.5° chimney divergence as optimal, achieving up to 3.66 times higher power output than the reference case. Further parametric analysis demonstrates that increasing chimney height and collector radius significantly enhances performance, while an ANN model with nine hidden neurons is successfully deployed to predict SCPP output with high accuracy. Mandal et al. [340] present an artificial intelligence-driven multi-objective optimisation (MOO) framework for the thermoeconomic design of solar chimney power plants. The framework integrates high-fidelity neural networks with genetic algorithms and CFD simulations to identify Pareto-optimal trade-offs between power, efficiency, and cost. Results indicate that collector parameters strongly influence cost, while chimney dimensions critically affect power and efficiency, with the optimised design achieving up to 4.51 times higher power and 5.73 times greater efficiency at only a 10% cost increase relative to the Manzanares reference plant. This surrogate-based optimisation approach substantially reduces computational demands compared to conventional CFD-driven methods and provides a scalable strategy for balancing thermodynamic performance with economic feasibility in solar chimney-based energy systems. Ghritlahre and Verma [341] compare ANN, group method of data handling (GMDH), and MLR models to predict the exergy efficiency of SAHs. The results reveal that the ANN model predicts with the highest accuracy, outperforming the other models. Ghritlahre et al. [342] analyse different input parameters using ANN models to predict the thermal performance of an SAH. Seven different ANN models are developed, with the ANN-II model yielding the best prediction results. This model utilises eight parameters, including mass flow rate, ambient temperature, wind speed, relative humidity, inlet, and average air temperatures, achieving an average relative error of 1.82% and a correlation coefficient of 0.99387. In their study, Ghritlahre et al. [343] employ an ANN to predict the thermal performance of an SAH. Experiments are conducted with two different SAH models (rough and smooth channels), developing seven different ANN models with

combinations of eight parameters. The most successful model is identified as the ANN-II model, exhibiting an average error of 1.82% and an R^2 value of 0.99387. Additionally, it is noted that the ANN model performs better than the GMDH model, with mass flow rate being the most sensitive parameter. In the study by Shafiey Dehaj et al. [344], the performance of a solar FP collector was modelled using an ANN and validated against experimental results. Two ANN models with five and four parameters are compared, with the four-parameter model yielding more accurate predictions. It was noted that ANN successfully predicted SFPC performance but could be further enhanced through GAs or PSO techniques. Ghritlahre and Prasad [345] predict the exergy efficiency of a rough SAH using three different ANN models (MLP, GRNN, RBF). Two different absorbers are employed in the experiments, collecting 210 datasets. The RBF model outperformed the other models, achieving the lowest error rates (RMSE and MAE) and the highest R^2 value. Rafique et al. [346] developed an SC model using LSTM networks to predict the energy output of an FPSC under various meteorological conditions. The LSTM model demonstrated superior performance compared to ANN models, achieving an R^2 value of 0.93. The LSTM model provides an efficient tool for forecasting SC performance in real-time applications. Rafique et al. [347] proposed a modified linear regression model for predicting solar PV energy output. The proposed model included parameters such as solar irradiation, temperature, and other atmospheric conditions, yielding promising results with a 97% prediction accuracy. This method enables efficient estimation of solar energy production, contributing to the optimisation of PV systems.

Accurate prediction of PV performance under general conditions is not possible using PV module manufacturers′ specifications. Therefore, the availability of an accurate and reliable solar PV system power prediction model is of vital importance [347].

Jathar et al. [348] review recent advancements in integrating AI and ML with PV systems and water treatment technologies. They emphasise the contributions of AI and ML in enhancing PV system performance, implementing predictive maintenance strategies, and enabling real-time monitoring, as well as in improving energy efficiency and environmental sustainability in water treatment processes. This review summarises innovative approaches in the fields of PV systems and water treatment for researchers and policymakers. Al-Dahidi et al. [349] aim to develop a model that predicts the efficiency of solar PV systems while considering variability. Their model uses historical operational conditions to predict the efficiency of test patterns and demonstrates superior results compared to traditional methods based on performance metrics such as MSE, R^2, and adjusted R^2. In a study by Demir [350], power forecasting methods for PV energy systems are examined, and three different forecasting models are developed using ML techniques. Evaluation with experimental data identifies the KNN algorithm as the most successful model with an 82% accuracy rate, while the SVM exhibits the lowest performance at 72%. Yadav et al. [351] predict the power output of a 2,680 kWp PV system in mountainous areas using temperature and irradiation data across various time intervals. The RBNN model shows the best performance over short time periods, while the kernel ridge regressor achieves optimal results for one-hour intervals. Kumaravel et al. [352] employ AI and ANN to predict the performance of PV systems in Oman. Experimental results show that the ANN controller effectively monitors system performance and is aligned with the values predicted by AI, with a two-layer feedforward neural network yielding the best prediction performance for small-scale PV systems. Bakht et al. [353] utilise ML models, including RF and SVM, to predict solar PV power output. Their research demonstrates the effectiveness of ML models in predicting solar PV power using data such as solar radiation, ambient temperature, humidity, wind speed, and precipitation. The results show that the RF model provides a slight advantage over the SVM model in evaluation criteria like R^2, RMSE, and MAE, underscoring the potential of ML methods to enhance the efficiency and reliability of solar energy systems. Keddouda et al. [354] conduct studies using 12 different ML algorithms to predict PV module temperature. Their findings reveal that the ANN model provides the highest accuracy with the lowest error, demonstrating superior performance compared to other models in the literature. Nassereddine et al. [355] focus on predicting PV power output and examine the advantages of transitioning from traditional methods to ML techniques. They compare the accuracy of PV power predictions using three different ML algorithms, with the ANN model standing out for its highest accuracy and lowest error rate, suggesting the future development of long-term forecasting models. Shivashankaraiah et al. [356] develop an innovative hybrid DNN model for short-term forecasting of solar PV power generation. Their study investigates the variables influencing PV power predictions, such as time, past PV power outcomes, and meteorological conditions, confirming that solar radiation is the most significant factor affecting PV output. Results forecast that the HDNN model outperforms other DL models in hourly forecasts. Liu et al. [357] propose a hybrid forecasting model integrating an optimised data clustering method with an AdaLSTM network to improve solar PV energy output prediction. Their study demonstrates that using cluster centres optimised by the iPSO algorithm enhances the prediction accuracy and reliability of the AdaLSTM model. Results indicate a significant reduction in RMSE values for the proposed model compared to other forecasting models, with a 10.75% decrease when compared to K-means clustering. Deshmukh and Mahobia [358] emphasise the necessity

for countries to shift towards renewable energy sources, particularly solar energy, due to climate change. They propose a hybrid model combining LSTM and CNN architectures, which is validated using a dataset obtained from Rabat, Morocco. The results reveal that the proposed hybrid structure surpasses traditional ML and DL algorithms in terms of prediction accuracy, sensitivity, and consistency. Jogunuri et al. [359] underscore the need for accurate forecasting of solar PV power for grid stability. Their study evaluates the performance of the random forest technique and other methods using weather data and power output measurements from a 20 kW installation, finding that the random forest technique provided a 49% and 50% accuracy increase for 15-min and 30-min forecasting horizons, respectively. Keddouda et al. [360] propose ANN and regression models to predict the power output of solar PV modules. Their study examines the impact of climatic conditions, such as solar radiation, ambient temperature, and module temperature, on PV power generation, and results indicate that the ANN model, particularly with the LM algorithm, achieves high accuracy. Additionally, the rational-power-law and power-law models successfully capture the system's nonlinear characteristics, demonstrating performance comparable to that of the ANN. Alghazo et al. [361] suggest predicting the performance of a solar tracking PV system installed in Zarqa, Jordan, using AI models. In their forecasts based on six months of experimental data, they test various classifiers using ANN, finding that the forest tree classifier provides the best prediction accuracy at 99.79% R^2, while linear regression yields the lowest accuracy. Banik and Biswas [362] develop a model for solar energy forecasting in Agartala, Tripura, India. Their study combines RF and CatBoost algorithms to predict solar radiation and PV energy generation, stating that the model is built on ten years of data and makes hourly predictions. The results show that the model achieves 6% higher accuracy than other methods. In the work by Scott et al. [363], various ML algorithms are compared for predicting PV system production at a university campus in Manchester. The findings indicate that the RF algorithm has the lowest error rate, emphasising the importance of data quality in enhancing the accuracy of algorithms. Wang et al. [364] introduce a hybrid DL approach that incorporates empirical mode decomposition and an improved LSTM model to enhance PV power generation predictions. Simulation results, based on data from Arizona, demonstrate that their proposed model predicts PV power with higher accuracy and stability than seven other models. Kumar et al. [365] investigate the performance of solar PV systems using real-time forecasting models based on historical data. They predict the output power of a semi-transparent PV system in Kovilpatti by using environmental factors for hourly, daily, and weekly predictions, achieving high accuracy with ELMAN, feedforward neural network (FFN), and GRN models. These results indicate consistent performance regardless of geographical location. Sharkawy [366] employs a multi-layer feedforward neural network (MLFFNN) to predict the power output of a solar PV power plant, addressing the continuous changes in climatic conditions. Conducted with six days of data, the first five days serve for training, while the sixth day is used to assess prediction accuracy. The results show that the MLFFNN effectively predicts PV power. Loukriz et al. [367] improve an ANN model to predict the output performance of PV modules under varying operating conditions to ensure the safe and stable operation of solar PV systems. Their model takes illumination values between 0 and 1000 W m^{-2} and temperature values between 0 and 100 °C as inputs, producing electrical characteristics as outputs. Results indicate that the ANN model exhibits a very low maximum power error of 0.02%, demonstrating accurate predictions of electrical properties. Tripathi et al. [368] utilise multiple regression, SVM regression, and Gaussian regression to predict the output power of solar PV panels. Their study finds that the multiple regression algorithm exhibits the best performance, achieving a low mean absolute error (MAE) of 0.04505. Rao et al. [369] assess the power output of solar PV systems using the ANN method, evaluating data such as daily global solar radiation, temperature, and sunshine duration. They define two weather conditions based on a "k" index derived from sunshine duration, resulting in MAPE values ranging from 1.888 to 10.599% for the developed ANN models.

Predictive modelling of AI-driven solar photovoltaic systems for weather data

Recently, there has been a significant surge in studies focussed on solar energy forecasting. Many researchers have concentrated on developing precise models using ML techniques. Among these, ANN has emerged as one of the most preferred methods for solar energy prediction [370]. In one of the previous sections, we briefly touched upon algorithms involved in forecasting for solar PV systems. Accordingly, this section will present a compilation of relevant literature on the application of these algorithms, highlighting system improvements achieved through their implementation.

ANN has demonstrated the ability to effectively analyse the complex and nonlinear relationships between solar energy generation and weather conditions [371]. However, it would be incorrect to claim that only ANN is successful in this area. For instance, in reference [372], they propose predicting PV power output using meteorological data with ANN, DNN, and LSTM models. In fact, LSTM outperformed ANN, achieving the best performance in power output prediction. Similarly, in the study by Ozdemir et al. [373], ANN and LSTM models are employed to forecast

solar irradiance for PV panels in Duzce, Turkey. Using collected meteorological and PV panel data, it was found that the LSTM model could provide more accurate predictions compared to ANN. Hu et al. [374] explore the impact of weather forecast data on PV energy production predictions and propose a model combining LSTM with a self-attention mechanism. This model improves prediction performance by up to 26.4% using weather forecast data. Zahraoui et al. [375] aim to predict PV power generation using a framework based on ANN and LSTM. This model, which utilises meteorological and electrical variables as inputs, successfully fills missing data and enhanced prediction accuracy, particularly when incorporating weather data. Nicoletti and Bevilacqua [376] develop two ANN models to predict PV electricity generation using weather forecast data (temperature, relative humidity, and wind speed). Sensitivity analysis reveals that hourly temperature trends are the most influential variable on prediction accuracy. While the first model uses more input data, the second model performs with similar accuracy using fewer inputs, indicating potential for energy communities to save energy and increase self-consumption. Lastly, Pang et al. [377] investigate the performance difference between DL algorithms, particularly RNN and ANN, for solar irradiance forecasting. Their study analyses the effects of data sampling frequency and moving window algorithms on achieving higher prediction accuracy. The results show that the RNN model outperforms the ANN model, offering 47% better normalised mean bias error (NMBE) and 26% better RMSE.

SVM, frequently applied in various studies, has emerged as one of the most widely used ML techniques for predicting the performance of solar PV systems based on weather data [378]. Niu et al. [379] focus on short-term power prediction of PV systems under sunny, cloudy, and rainy weather conditions. In their research, an optimised dragonfly algorithm (IDA) is employed to enhance SVM using data such as temperature, relative humidity, and solar radiation. The results demonstrate that the IDA-SVM model achieves high prediction accuracy across different weather conditions, thereby improving the reliability of energy systems by enhancing PV energy integration. Similarly, Saxena et al. [380] highlight the impact of uncertainties in weather conditions on PV power prediction. Their study develops a hybrid model, using KNN and SVM to forecast PV power generation based on weather data. This hybrid model, tested on the Jodhpur dataset, yields high accuracy in PV energy generation forecasts. Moreover, Raj et al. [381] propose ensemble-based ML models, including gradient boosting machines (GBM) and RF algorithms, to forecast power generation from six different solar PV systems. These models, based on three years of performance data from a 1.2 MW grid-connected solar PV plant, utilised input features selected through principal component analysis (PCA). The models' accuracy was compared to algorithms such as KNN and SVM, with the results indicating that SVM and decision tree-based ensemble algorithms outperformed the KNN model. Wu et al. [382] propose three hybrid models, based on ACO, cuckoo search, and GWO, to optimise the SVM algorithm. These models are used to predict global solar radiation in four different climate regions of China (temperate continental, Plateau Mountain, temperate monsoon, and subtropical monsoon climates). The findings reveal that, among stand-alone models, SVM provides the highest accuracy for Rs prediction, while the hybrid models achieve even higher accuracy than the stand-alone SVM model. In particular, the GWO-SVM and CS-SVM models exhibit superior performance in solar radiation forecasts compared to other models. Jobayer et al. [383] underscore the importance of predicting PV system performance, systematically reviewing ML-based PV parameter prediction studies published between 2020 and 2022. Their review concludes that ML algorithms provide faster and more reliable predictions than conventional methods. Specifically, ANN is found to be the most popular ML method (32.55%), followed by random vector functional link models (13.95%) and SVMs (9.30%). Agarwal et al. [384] emphasise that the variability in weather conditions affects PV output, making accurate predictions critical for grid integration. They develop a model based on weather classification to predict solar PV output, utilising SVM for weather classification (sunny, rainy, cloudy), with an ANN model developed for each weather condition. This combination achieves higher accuracy in PV output prediction, with a lower error rate (1.53%) compared to previous studies. Liu et al. [385] propose an improved whale optimisation algorithm to optimise SVM in order to reduce uncertainty in PV energy production forecasts. Their study highlights that SVM requires less training data and delivers high prediction accuracy under various weather conditions. Additionally, the model's performance is improved by reducing data noise through a wavelet soft-thresholding method. In another comparative study, de Freitas Viscondi et al. [386] analyse ML algorithms for PV energy production forecasts. Their findings underscore that, particularly in São Paulo, SVM provides the best prediction performance, achieving the lowest RMSE. Notably, when all meteorological parameters are used, SVM offers the highest accuracy without compromising fast training times. Chuluunsaikhan et al. [387] use ML models to predict the power output of solar panels based on weather and air pollution features. Their study reveals that methods such as SVR yield successful results when forecasting with weather data, though air pollution data resulted in lower accuracy. Meanwhile, Baltaci et al. [388] employ temperature and irradiance, key meteorological factors reflecting weather impacts, when predicting PV power output. Despite not addressing other weather variables such as dust, humidity, wind speed, and precipitation, their study shows

that the SVM method provides higher accuracy compared to ANN when forecasting with temperature and irradiance data. In another study, Kuriakose et al. [389] compare ML algorithms for predicting the power output of solar PV systems. Considering weather parameters such as wind speed, humidity, irradiance, and temperature, their results indicate that, while the SVM model delivers comparable accuracy to ANN, ANN offers higher prediction accuracy overall. SVM, however, contributes to more accurate forecasts of power output when based on weather data. Sabzehgar et al. [390] develop a neural network (NN)-based numerical weather prediction model for solar energy forecasting. By comparing the accuracy of this NN-based model with alternative methods such as SVM, their results show that while SVM provides significant accuracy, the NN-based model offers superior accuracy and efficiency. Finally, Visser et al. [391] explore the influence of weather data on the performance of ML models, including SVM, in solar energy forecasting. Evaluating the predictive accuracy of SVM for 152 PV systems, both individually and collectively, the results highlight the critical role of weather variables, including surface solar radiation, cloud cover, and temperature, on forecast performance. Similarly, Wang et al. [392] develop a weather classification model for PV power forecasting. Examining the performance of traditional ML models like SVM, they demonstrate that finer weather classification significantly improves the accuracy of PV power predictions. The model, enhanced through data augmentation using GAN and CNN, improves SVM's forecasting success. A summary table for current literature studies has also been created, as seen in Table 4.

The forecasting models in the table aim to provide more accurate predictions about energy production by considering various meteorological data, the characteristics of PV system components, and environmental factors. The accuracy of these models plays a critical role in system design and investment decisions. Additionally, they provide important data for energy management and demand forecasting, facilitating the integration of energy networks. In the coming years, it is expected that further development of forecasting models will be achieved through the use of advanced technologies such as AI and ML. These innovations will not only enhance efficiency but also strengthen the real-time performance monitoring capabilities of the systems. Therefore, the future development of forecasting models for PV systems will contribute to broader applications in the energy sector and energy independence.

Hybrid energy applications based on AI

Hybrid energy systems, which are utilised for the purpose of energy provision or power generation, typically integrate both renewable and fossil fuel sources [393]. These systems not only aim to enhance energy efficiency but also strive to ensure reliability while minimising dependence on a single energy source. By combining renewable resources such as wind, solar, or biomass with traditional energy methods like diesel generators or even other renewable sources, hybrid solutions offer a more adaptable and balanced energy framework [394]. However, the implementation of hybrid energy systems encounters various challenges, as highlighted in the literature. Among these, technical and economic issues arise, including the need to ensure continuity in energy production, address efficiency losses, achieve cost-effectiveness, and tackle challenges related to storage and distribution, as well as the integration of systems [395]. Consequently, common hybrid approaches are designed to address these problems through the amalgamation of diverse energy sources. To further enhance the efficiency of these hybrid systems and effectively resolve existing issues, AI applications are increasingly being employed. Notably, AI provides significant solutions, particularly in areas including energy production and consumption forecasting, system optimisation, energy storage management, and fault diagnosis. By implementing AI-driven methodologies, we can effectively monitor hybrid system performance, ensuring the most efficient use of energy resources. As a result, energy losses can be minimised, facilitating more balanced and sustainable energy management.

Koholé et al. [396] optimise a hybrid energy system that combines PV, wind, diesel, and fuel cells to meet varying load demands in Figuil, Cameroon. Their study compares the techno-economic viability of this configuration with others, revealing that the PV/wind/diesel/fuel cell (FC) system emerges as the most cost-effective and environmentally sustainable option. This hybrid system significantly reduces CO2 emissions while ensuring a reliable energy supply. Additionally, the researchers evaluate various evolutionary algorithms, identifying the variable population size (VPS) algorithm as the most effective for system design. In a related study, Sivaraman et al. [397] focus on designing a grid-connected hybrid energy system that aims to enhance performance, inject power into the grid, and minimise greenhouse gas emissions. Utilising the non-dominated sorting genetic algorithm II (NSGA-II), they optimise the system's dimensions and explore different operational scenarios. The findings indicate that this method improves energy resource utilisation, diminishes output power fluctuations, and requires fewer batteries for storage, thereby enhancing both the technical and economic performance of the hybrid

Table 4 Literature summary about the predictive modelling of solar photovoltaic systems for weather data with AI

Authors	Publication year	Research Topic	Models	Outcomes
Ozdemir et al. [373]	2022	Prediction of Solar Irradiation for PV Panels	ANN, LSTM	The LSTM model achieved higher accuracy compared to the ANN
Hu et al. [374]	2024	Impact of Weather Forecast Data on PV Energy Production Predictions	LSTM, self-attention mechanism	The model improved the prediction performance by 26.4%
Zahraoui et al. [375]	2024	Predicting PV Power Generation	ANN, LSTM	Increased forecasting accuracy by considering weather data
Nicoletti and Bevilacqua [376]	2024	Prediction of PV Electricity Generation	ANN	The most influential variable on the accuracy of hourly temperature trend forecasting
Pang et al. [377]	2020	Comparison of DL Algorithms for Solar Radiation Prediction	RNN, ANN	The RNN model provided a 47% better NMBE and a 26% better RMSE compared to the ANN model
Niu et al. [362]	2024	Short-Term Power Forecasting of PV Systems	IDA, SVM	The IDA-SVM model provided high prediction accuracy under different weather conditions
Saxena et al. [379]	2024	The Influence of Weather Uncertainties on PV Power Forecasting	KNN, SVM	The hybrid model achieved high-accuracy
Scott et al. [363]	2023	PV System Performance Forecasting	Various ML algorithms, SVM	SVM provided effective predictions with low average error rates
Raj et al. [381]	2023	Ensemble-Based ML Models	GBM, RF	The models achieved higher accuracy compared to other algorithms
Wu et al. [382]	2023	Optimising the SVM Algorithm	Hybrid models with ACO, CS, and GWO	Hybrid models achieved higher accuracy compared to the SVM model
Jobayer et al. [383]	2023	ML-based PV Parameter Prediction Studies	Various ML algorithms	ML algorithms provided faster and more reliable predictions compared to traditional methods
Agarwal et al. [384]	2022	The Impact of Weather-Dependent Variability on PV Output	Weather classification, SVM	Achieved higher accuracy in PV output forecasting
Liu et al. [385]	2021	Reducing Uncertainty in PV Energy Production Forecasts	Improved whale algorithm, SVM	SVM required less training data and achieved high prediction accuracy
de Freitas Viscondi et al. [386]	2021	Performance of ML Algorithms in PV Energy Production Forecasting	Various ML algorithms	SVM provided the best prediction performance with the lowest RMSE value
Chuluunsaikhan et al. [387]	2021	Forecasting the Power Output of Solar Panels	SVR	Successful results are achieved with weather data
Baltaci et al. [388]	2021	Forecasting the Power Output of PV Systems	SVM, ANN	SVM achieved higher accuracy in predictions
Kuriakose et al. [389]	2020	Forecasting the Power Output of Solar PV Systems	SVM, ANN	SVM and ANN provided similar accuracy; however, ANN offered higher accuracy
Sabzehgar et al. [390]	2020	Development of an NN-based Model for Solar Energy Forecasting	NN, SVM	The NN-based model offered higher accuracy and efficiency
Visser et al. [391]	2019	The Impact of Weather Data on SVM	SVM	The impact of weather variables on forecasting performance has been highlighted

Table 4 (continued)

Authors	Publication year	Research Topic	Models	Outcomes
Wang et al. [392]	2019	Development of a Weather Classification Model	SVM	More refined weather classification can improve the accuracy of PV power forecasting

system. Bacha et al. [398] examine the optimal sizing of a hybrid energy system comprising photovoltaic panels, wind turbines, energy storage, and diesel generators using AI methods. In their study, they employ the SFS and SOS algorithms, with the SFS algorithm demonstrating superior performance in meeting energy demands at the lowest cost. Ahmed et al. [399] investigate the integration of PV and wind energy into a hybrid energy system, concentrating on maintaining a stable DC bus voltage. They analyse the performance of this hybrid system through MATLAB simulations, comparing it to fossil fuel-based energy generation methods. Their work underscores the potential of hybrid systems in addressing Pakistan's energy challenges. Rusilowati et al. [400] explore the role of AI in the development of renewable energy sources and the enhancement of energy efficiency. Their research reveals that AI contributes to operational efficiency by optimising the placement of solar panels, facilitating predictive maintenance for wind turbines, and improving fermentation processes in biogas production, thereby reducing energy waste. The study highlights the significant contributions of AI in promoting efficiency and sustainability within the energy sector. Soni et al. [401] aim to analyse and optimise the performance of grid-connected solar PV-fuel cell hybrid power systems through an AI-assisted approach. Their findings demonstrate how AI can assist in decision-making processes, enhance system performance, and improve energy efficiency while ensuring financial sustainability. The results indicate that this hybrid system can fulfil 74% of the energy needs of a university building, thereby reducing energy costs and minimising CO2 emissions, thus enhancing environmental sustainability. Belboul et al. [402] develop an improved salp swarm algorithm to optimise an isolated microgrid consisting of solar panels, wind turbines, batteries, and diesel generators. Their proposed method achieves significant results, determining energy costs at \$0.2109/kWh, a net present cost (NPC) of \$376,063.8, a 4% probability of energy loss, and annual CO2 emissions of 12.4457 tonnes, showcasing the highest cost-effectiveness of this hybrid system configuration. In the work of Hai et al. [403], the economic and environmental impacts of a hybrid energy system for a healthcare centre in Rijal Almaa, Saudi Arabia, are assessed. Analyses conducted using HOMER software identify a configuration comprising 360 batteries, a 25 kW diesel generator, a 2 kW wind turbine, and 33.3 kW solar panels as the most efficient system, with an NPC of \$109,307 and a cost of energy (COE) of \$0.103/kWh. Additionally, the system achieves a renewable energy share of 84.7%, resulting in annual CO2 emissions of 10,825 kg and an excess electricity production rate of 8.81%, indicating a positive environmental impact. Barua et al. [404] introduce an advanced optimisation algorithm for managing solar/wind/diesel microgrid energy for remote communities. Their developed Lévy arithmetic algorithm addresses the economic emission distribution problem, aimed at minimising energy costs and emission levels. The analyses reveal that this algorithm provides better cost savings compared to other meta-heuristic optimisation methods and that integrating solar and wind energy reduces dependence on conventional fuel sources and decreases carbon emissions. Chen and Zhang [405] focus on the technical, economic, and environmental evaluations of hybrid energy systems (PV and diesel generators) for remote islands. They employ a modified swarm intelligence algorithm to minimise energy costs and CO_2 emissions, comparing results with standard swarm intelligence and simple simulated annealing methods. Their findings indicate that the most cost-effective and low-carbon configuration comprises the hybridisation of solar energy, diesel generators, and battery energy storage systems, with energy costs ranging from \$0.2845 to \$0.6492/kWh. Al Hajri et al. [406] develop a hybrid renewable energy system integrating solar, wind, and hydrogen fuel cells for IoT applications. This system, which optimises energy production using ML algorithms, offers a continuous energy supply and promotes sustainable energy solutions. Future studies are recommended to optimise the system for varying climatic conditions. Elkadeem et al. [407] present the design and control framework of a large-scale grid-connected hybrid solar/wind energy system for the Sokhna Industrial Zone in Egypt. The results obtained demonstrate that the integration of solar and wind systems offers both sustainability and economic advantages. Furthermore, the proposed control model enhances the dynamic behaviour and power quality of the system, providing an effective tool for renewable energy projects. Sharma [408] introduces a new control strategy involving the integration of solar and wind energy into a central DC microgrid for electric vehicle charging stations. This study aims to ensure that sufficient energy is provided to charging stations through the effective

use of solar and wind resources. Simulations are conducted using PSIM 9.0 software, and the resulting data are presented. The findings highlight that while solar PV technology fosters sustainability, there is a need for enhanced efficiency. Merino and Castro [409] create an optimisation model that employs GAs for a hybrid solar–wind microgrid tailored to fulfil the energy requirements of 2000 households in Chile. This project incorporates a 2.5 kWp solar panel along with a 12.3 MW wind turbine, which collectively provides around 85% of the necessary energy from renewable sources. Economic assessments reveal an NPV of €25 million and a levelised cost of €37/MWh, underscoring the vital role of microgrids in advancing sustainable energy solutions. Ertas et al. [410] manage the power distribution in a hybrid power plant featuring a 3.125 MVA diesel generator paired with a 2 MW solar panel. They utilise ANN to forecast the power generated by the solar panel, which allows for the diesel generator's operation to be adjusted based on these predictions, effectively reducing fuel consumption. Simulation outcomes indicate that solar energy production is accurately controlled over time, and this management strategy leads to decreased carbon dioxide emissions. Gurumoorthi et al. [411] introduce a novel hybrid model designed to tackle the optimal power flow problem inherent in renewable energy systems integration. This model adeptly addresses uncertainties within energy systems while striving to reduce fuel costs and minimise power losses by fusing deep reinforcement learning (DRL) with quantum-inspired genetic algorithms. Experimental findings suggest that the proposed model yields benefits, including reduced fuel expenses ($620.50) and diminished power losses (1.85 MW), although its real-time application faces challenges due to significant computational demands. Kenfack et al. [412] carry out a sustainability and cost evaluation of seven distinct PV/T hybrid systems, devising an optimal selection methodology through GAs and multi-objective PSO. Their research highlights an energy cost of $0.1/kWh, with a payback period under three years, indicating that liquid cooling systems outperform air systems in heat extraction efficiency. Moreover, the bifacial model demonstrates advantages regarding the energy-environmental balance. Amoussou et al. [413] investigate the optimisation of an autonomous hybrid renewable energy system (HRES) intended for residential use in Buea, Cameroon. By employing the Crayfish optimisation algorithm (COA), they compare the costs associated with PV–battery and PV–battery–diesel systems, revealing that COA results in lower energy costs across both configurations than other algorithms. Consequently, the PV–battery–diesel setup emerges as the most efficient with superior cost performance. Mittal and Karuna [414] explore the management of wind–solar hybrid microgrids through swarm intelligence algorithms, optimising the intricate interplayamong environmental conditions, energy demand, and battery storage. Their research indicates that the PSO algorithm outperforms its counterparts, achieving an 80% renewable energy penetration rate alongside a 15% reduction in costs. Vadlamudi's study [415] suggests a method leveraging stationary discrete wavelet transformation and a hybrid LSTM-CNN classifier for detecting islanding faults within wind and solar energy systems. This approach, enhanced by AVOA, achieves an impressive accuracy rate of 99.92% across varying noise levels. Zhu et al. [416] design and optimise a hybrid renewable energy system comprising a diesel generator, PV panels, and a battery storage system. Their study employs an MOO strategy through a custom algorithm focussing on minimising costs, reducing carbon emissions, and ensuring load satisfaction. The results demonstrate that the proposed method enhances energy production while decreasing carbon emissions by 8.52%. Nkalo et al. [417] create a modified MOPSO algorithm aimed at the optimal sizing of a solar–wind–battery hybrid renewable energy system for a rural community in Rivers State, Nigeria. Their work prioritises minimising energy costs while also lowering the likelihood of energy supply loss. Findings reveal that modified MOPSO effectively reduces LPSP, albeit at a higher levelised cost of energy (LCOE), thus increasing system reliability. Additionally, the proposed system meets energy demand under favourable weather conditions; however, adverse conditions necessitate more frequent battery discharges. Vani et al. [418] devise an innovative neural network MPPT system for a DC microgrid that integrates PV and wind energy systems. This system optimises energy harvesting and manages battery storage efficiently, adapting to fluctuations in environmental conditions and load demands. Simulation results indicate that the IDNN system achieves lower errors, reduced total harmonic distortion, and diminished computational complexity compared to conventional methods. Soni et al. [419] present a study that examines the integration of grid-connected solar PV and FC hybrid power systems, addressing the rising demand for clean and sustainable energy solutions. They conduct a comparative techno-economic analysis using AI techniques, such as GA and PSO, to evaluate the performance, economic feasibility, and environmental impact of these hybrid systems. Results demonstrate that the hybrid system achieves a LCOE than standalone solar PV systems, attributed to the FC's capability to supply power during low sunlight conditions, including at night and on overcast days. Imam Rahmani et al. [420] develop an AI-driven framework that integrates the GWO and radial basis function neural network for power forecasting in hybrid renewable energy systems. This methodology optimises model parameters to achieve more precise power forecasts, showcasing adaptability to various operational conditions. Performance evaluations reveal that this framework offers superior forecasting capabilities, marked by lower error rates relative to existing models. Roy et al. [421]

implement ML techniques to assess the technical, economic, and environmental performance of hybrid renewable energy systems on Henry Island. They analyse four distinct system combinations via ML, with the bilayered neural network model yielding the highest accuracy in LCOE predictions. Their findings accentuate the economic and environmental ramifications of system combinations featuring wind turbines, photovoltaic panels, biogas generators, and batteries. Hazarathaiah and Reddy [422] underscore the crucial role of battery storage systems in integrated wind–PV-based stand-alone electric power supply systems. The authors innovate control solutions for DC–DC converters and inverters, employing LSTM-based ANN control methods to enhance power quality. Their objective is to ensure a continuous and reliable energy supply despite fluctuations in loads, wind speeds, and solar irradiation conditions.

Hai et al. [423] propose a system that enhances the integration of renewable energy into local energy grids by employing AI techniques for optimisation. They conduct thermodynamic, environmental, and exergo-economic assessments to identify optimal operational conditions through MOO methods implemented in MATLAB. The use of AI algorithms aims to improve both the energy efficiency and cost-effectiveness of the system, ultimately focussing on reducing exergy losses from the combined utilisation of solar and biomass resources. Irshad et al. [424] evaluate hybrid renewable energy systems that integrate solar energy and biomass sources, applying multi-objective genetic algorithms (MOGA) for optimisation. Their findings indicate that the pyrolysis system represents the most cost-effective option, achieving the highest green energy ratio and the lowest harmful emissions compared to alternative biomass technologies. Nouri et al. [425] introduce an innovative polygeneration system that generates electricity, heat, hot air, oxygen, and ammonia by integrating solar and biomass energies. This system undergoes evaluation using a multi-objective GWO algorithm, revealing total energy and exergy efficiencies of 70.68% and 48.71%, respectively. Modu et al. [426] design a hybrid renewable energy system that combines solar, wind, biomass, fuel cells, and battery storage to meet the energy needs of a rural community in north-eastern Nigeria. They optimise the sizing of system components using the Levy flight-salp swarm algorithm (LF-SSA), resulting in a significant reduction in annual system costs. Their results demonstrate that the proposed energy management system effectively establishes an eco-friendly and cost-efficient energy infrastructure. In a separate study, Modu et al. [427] develop an HRES that incorporates solar, biomass, hydrogen fuel cells, and battery storage, optimising component sizing through the Levy and sine–cosine operator-based salp swarm algorithm (LSC-SSA). Their research shows that LSC-SSA offers substantial cost savings compared to traditional salp swarm and GAs, achieving the lowest energy costs. Ghiyami et al. [428] introduce a continuous multi-production hybrid energy system based on biomass and liquefied natural gas (LNG), implementing carbon capture, utilisation, and storage methods. Simulations and ML algorithms demonstrate the potential to produce 130 kg/h of hydrogen, 375 kWh of energy, and 2600 kg/h of precipitated calcium carbonate using biomass. They develop a desktop application that enables users to easily calculate the effects of various variables while concentrating on the environmental compliance and cost-effectiveness of the proposed system. Fang et al. [429] assert that an integrated system combining solar and biomass energy technologies serves as an effective solution for addressing greenhouse gas emissions and energy production costs. In their study, they create a model using an LSTM-RNN algorithm to generate a dataset of 280,000 for various scenarios, analysing the system's carbon savings and economic potential through an MOO approach. Their findings reveal that the most favourable scenario results in a CO2-eq savings of 415,960 tonnes and an NPV of €4.298 million over 30 years. Sakib et al. [430] propose a hybrid renewable energy system that incorporates solar, biomass, and battery components in the Mymensingh region of Bangladesh, optimising the system's sizing and costs. They address uncertainties in solar radiation and load demand through a probabilistic scenario-based analysis. The dandelion optimisation algorithm (DO) demonstrates an 18.4% reduction in simulation time and a 3.2% cost reduction compared to two other algorithms. Their results provide a cost-effective forecast for local HRES development, emphasising the importance of considering uncertain parameters in off-grid areas of Bangladesh. Irshad et al. [431] design a 100% renewable solar/biomass-based energy system without employing energy storage devices. The configuration, optimised with an MOGA, estimates an NPC of $118,942, a COE of $0.02/kWh, and a total system cost of $892,892. This proposed system effectively meets load demands without the need for energy storage technologies, preventing 729.5 tonnes of CO_2 emissions by annually consuming 704.81 tonnes of wheat straw. Ishraque et al. [432] present the operation and evaluation of a hybrid microgrid that integrates solar, wind, biomass, hydro, and electrolyser resources, employing a model predictive deterministic algorithm to optimise load variations. Their research illustrates the hybrid microgrid's potential to provide a reliable power source and enhance performance through smart control strategies across various load scenarios. The results indicate that while the grid-connected mode offers lower energy costs, the isolated mode performs better in terms of minimising costs and emissions. Liu et al. [433] propose a life cycle assessment (LCA)-based MOO model for the optimal design of a hybrid biomass and solar energy system. The study analyses the environmental impact, renewable energy contribution, and economic benefits of the hybrid system,

showing that optimised performance achieves an environmental impact load reduction rate of 46.03%, a renewable energy contribution rate (RECP) of 92.73%, and an annual total cost savings rate of 35.75%.

Muller et al. [434] assess the techno-economic feasibility of hybrid renewable energy systems using HOMER simulations, identifying solar PV combined with wind turbines as the most cost-effective configuration for an academic campus in India. The optimised system, comprising a 3007 PV array, two 1.5 MW wind turbines, and a 1927 kW converter, reduces annual CO_2 emissions by over 42% and lowers operational costs by 40.8% compared to the existing grid, achieving a payback period of just over ten years. The study highlights that such hybrid systems not only provide reliable on-site electricity production but also offer substantial economic and environmental benefits, with potential replicability in regions with similar climatic conditions. Sunil and Sharma [435] propose AI-based control strategies for a hybrid renewable energy system comprising solar PV and wind turbines. These strategies reduce voltage and frequency stress alongside energy storage systems, enhancing power quality and increasing system reliability. Simulation results demonstrate significant improvements in the power quality and stability of the hybrid system. Emrani et al. [436] explore how hybrid energy systems offer environmentally friendly and economically viable solutions to meet global energy demands. They present a model integrating wind turbines (WT) and PV systems with a hybrid energy storage system that combines mechanical gravitational energy storage and electrochemical battery storage systems. Optimisation and forecasting models balance the reliability and cost of renewable energy sources. Salazar-Caceres et al. [437] develop a method focussed on uncertainty analysis to assess the technical feasibility of solar–wind hybrid energy systems. This approach, based on local weather data, aims to enhance the accuracy of system design by utilising critical metrics such as performance ratio, capacity factor, and performance returns. The results highlight the limited technological applicability of wind energy while emphasising that accurately accounting for uncertainties can improve the efficiency and reliability of renewable energy systems. Atawi et al. [438] investigate the optimisation of hybrid energy systems that include wind and solar energy using the multi-objective African vultures optimisation algorithm (MOAVOA). Conducting a comprehensive case study in Tabuk, Saudi Arabia, they evaluate system designs in both grid-connected and stand-alone modes. They compare MOAVOA with other algorithms, such as NSGA-II and MOPSO, analysing the Pareto front to identify optimal system designs. Duan et al. [439] optimise a hybrid microgrid that includes PV, WT, and battery energy storage (BES) in their study. They aim to minimise energy losses, voltage fluctuations, and purchased energy using the multi-objective Kepler optimisation algorithm. Their work examines how battery discharge depth affects energy losses and voltage deviations in the grid across three scenarios based on actual and predicted data. The results indicate that optimisation based on forecast data increases annual energy losses and costs. Ramakrishna et al. [440] propose a hybrid controller based on the gradient boosting decision tree and walrus optimisation algorithm (WOA) to optimise the performance of hybrid renewable energy systems that rely on solar and wind energy sources. Despite the variable system conditions in HRES, MPPT technology ensures the tracking of the maximum power point. The proposed controller is tested on MATLAB/SIMULINK and compared with existing methods. The results demonstrate that power generation fluctuates according to system conditions and that the proposed controller successfully tracks the optimal duty cycle. Donald and Bo [441] conduct a comprehensive study on the modelling, control, and optimisation of hybrid solar–wind energy conversion systems. Their research develops control algorithms to optimise the system in response to variations in environmental conditions, showing that the proposed model increases storage efficiency to 90% and reliability to 98.63%. Serir et al. [442] focus on the significance of energy management control in hybrid energy systems that include PV solar and wind resources alongside a battery bank. The developed smart management system utilises a fuzzy logic controller (FLC) to precisely monitor the battery status across five different levels, ensuring that the battery remains above 60% during charging and below 27% during discharging. Ma et al. [443] highlight the importance of energy storage in addressing continuity issues in hybrid energy generation systems that involve wind, photovoltaic, and storage. The nine scenarios produced using the WGAN-GP model indicate that energy storage can reduce renewable energy interruptions by between 3.6 and 14.7%. Said [444] emphasises that hybrid renewable energy systems, formed by the combination of WT and PV systems, provide a cost-effective solution. The system, optimised using the RIME algorithm, meets energy demand with 98 PV arrays, 187 wind turbines, and 414 batteries, ensuring a 0% loss probability and low energy costs (COE). Ultimately, the study states that the RIME algorithm achieves the lowest costs and loss probability compared to other methods like PSO and CSA. Ghanbari et al. [445] address the design and optimisation of hybrid renewable energy systems that operate using wind and solar energy to meet drinking water and electricity demands. In their research, various algorithms are employed to determine the best system sizing, minimising costs and enhancing reliability, revealing that PSO and MOPSO algorithms perform best. Chegari et al. [446] underscore the significance of local energy production systems in positive building design, proposing a new energy management system focussed on optimising hybrid

renewable energy systems. Their research develops a hybrid local energy production system that includes solar photovoltaic, wind, pumped storage, and battery storage, modelling and calculating processes in the TRNSYS environment, demonstrating that the proposed system achieves 39% more energy self-sufficiency to meet energy demand.

Farajollahi et al. [447] propose a novel energy system in their study, aimed at enhancing the efficiency of hybrid solar-geothermal energy systems due to the lack of solar thermal energy storage. The research employs an optimisation process utilising ANN and RSM with five independent variables. The results indicate that under specific conditions, certain parameter values achieve a thermal efficiency of 30.47% and a cost of $13.04/GJ. In the work by Yang et al. [448], an innovative approach is presented to improve the energy efficiency and cost-effectiveness of geo-solar systems, which consist of ground source heat pumps and solar thermal collectors. This study integrates ANN and multiple optimisation methods to optimise system parameters, yielding results that show an exergy efficiency of 22.02% and turbine power of 1963 kW. Farahani and Farahani [449] investigate the freshwater production of a humidity-dehumidification-based water purification system across five different cities in Iran, using TRNSYS software. The study evaluates the freshwater efficiency obtained by using various heat sources, such as solar energy and geothermal energy, along with three different types of collectors. The findings reveal that the highest freshwater production occurs in Hamedan city when utilising a combination of GE and compound parabolic collectors. The results also demonstrate that freshwater efficiency can be successfully predicted using ML and DL algorithms, with the Gaussian process method showing a slight advantage over the LSTM method. Elkholy et al. [450] present a new design and flexible energy management strategy for microgrids in this study. The developed intelligent energy management system optimises energy management by integrating a hydrogen-battery hybrid energy storage system alongside renewable energy sources. Experimental tests highlight the role of hydrogen in energy management, showing a reduction of 1.287% in operational costs, and demonstrate the effectiveness of the proposed multi-objective algorithm compared to other algorithms. Liu et al. [451] emphasise the significance of accurate ageing assessment in determining the optimal operating mode for regional wind-hydrogen integrated energy systems (RWHIES). Their work introduces a wind turbine power forecasting method (WPITC) using wind physics information and topographical characteristics. The developed model aims to maximise production capacity at minimum cost by assessing the ageing condition and solves the optimisation problem using a multi-objective dragonfly algorithm (IMODA). The findings indicate that the proposed approach enhances both the accuracy and speed of cost and capacity optimisation for RWHIES.

Future prospects

In this section, a SWOT analysis is applied to clarify the strengths, weaknesses, opportunities, and threats associated with the integration of AI in solar thermal and PV systems. The goal of this analysis is to offer a comprehensive understanding of the various aspects influencing AI-based solar energy research, helping to identify areas of growth while acknowledging the challenges and risks that need to be addressed in future advancements.

The strengths of AI-enhanced solar energy systems lie primarily in their ability to significantly improve efficiency across a range of applications. AI techniques, such as ML, ANN, and DL, have been successfully employed in areas like prediction control, fault detection, and performance optimisation. These advancements allow for more accurate energy production predictions and better overall system performance, contributing to the reliability and scalability of solar thermal and PV systems. Moreover, these systems have a wide range of applications, including electricity generation, water heating, industrial processes, and agriculture, demonstrating their versatility and potential to reshape energy sectors globally.

The increasing global demand for renewable energy further enhances the opportunities for AI-enabled solar systems. As environmental awareness grows and governments introduce policies favouring clean energy, the adoption of AI-integrated solar systems could become more widespread. Additionally, advancements in AI algorithms and nanotechnology are expected to offer more efficient and sustainable solutions, addressing both energy and environmental concerns. These developments present substantial opportunities for future research and commercialisation of AI-driven solar energy systems.

Despite the numerous advantages, the integration of AI into solar energy systems comes with its own set of weaknesses and challenges. The initial costs of implementing AI and nanotechnology into research systems are significantly higher, making them less accessible for certain regions or institutions. Moreover, these advanced systems require highly specialised technical expertise for installation and maintenance, creating a barrier to entry for widespread adoption, especially in regions lacking the necessary infrastructure and human capital. Another critical weakness is the vulnerability of solar systems to environmental conditions. Solar energy output can be negatively impacted by factors such as cloud cover or shading, making it dependent on AI's ability to manage and mitigate these variables effectively. Additionally, AI-based systems require large datasets to operate accurately, and the quality of the data used can directly impact the system's performance.

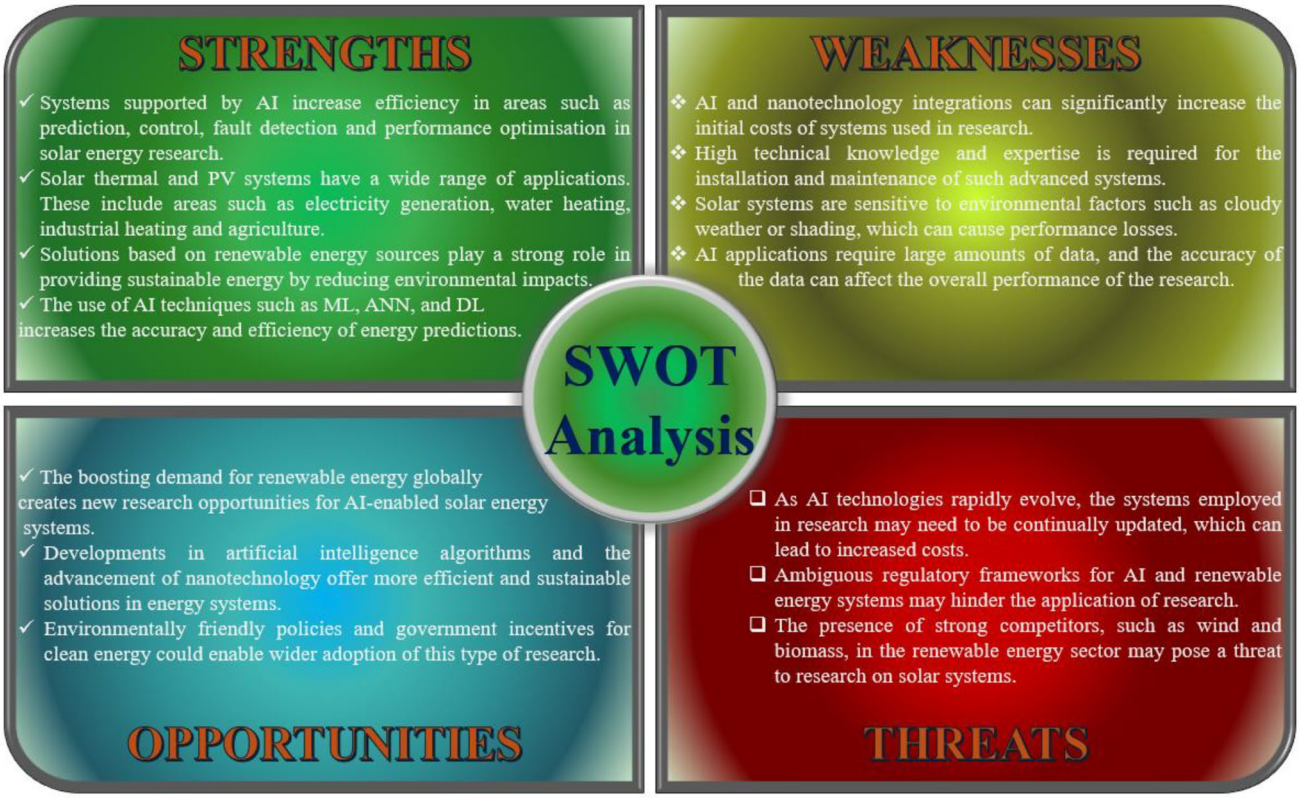

Fig. 5 SWOT analysis providing a summative overview of the usage of AI applications in solar thermal and PV systems

From a broader perspective, the rapid evolution of AI technologies presents a significant threat in terms of costs. As AI advances, research systems may need continual updates to keep pace with the latest developments, which can increase operational costs. Moreover, the regulatory landscape for AI and renewable energy remains ambiguous, potentially stifling the broader application of AI-based solutions in solar research. Competition from other renewable energy sources, such as wind and biomass, also poses a threat to the prominence of solar-based systems in the renewable energy sector.

The future of AI-integrated solar energy systems holds great promise. The ability of AI to enhance efficiency, accuracy, and performance across a wide range of applications positions solar thermal and PV systems as essential tools in the transition to sustainable energy. However, several challenges must be overcome, including high initial costs, technical complexity, and the ongoing need for regulatory clarity. One critical point to consider moving forward is the balance between innovation and accessibility. While advanced AI-driven solutions offer tremendous potential, ensuring that these technologies can be widely adopted across diverse geographic and economic regions will be key to their long-term success. Similarly, maintaining a focus on developing robust regulatory frameworks that support both AI technologies and renewable energy will be essential to minimising the risks associated with these evolving systems. In conclusion, while there are clear obstacles, the ongoing research and development in AI-enabled solar systems could play a pivotal role in shaping the future of sustainable energy, provided that the industry continues to address the weaknesses and threats identified in this analysis. By leveraging the opportunities presented by global demand for clean energy and advancements in AI, solar energy research has the potential to make a lasting and transformative impact on energy systems worldwide. Figure 5 provides a summary of all that has been written above.

Conclusions

Solar energy serves as one of the cornerstones of sustainable energy production and plays a crucial role in meeting rising energy demands. STSs and solar PV systems enable the effective utilisation of this energy source. However, there is a need for innovative solutions to enhance the performance of these systems, maximise energy efficiency, and reduce environmental impacts. In this context, AI applications hold

significant potential in the management and optimisation of solar energy systems. AI employs data analysis and modelling techniques to improve energy production processes and support decision-making, thereby increasing the efficiency of solar energy. Moreover, hybrid energy applications and innovative energy solutions become more effective with the integration of AI. This study comprehensively examines the role of AI in solar energy systems and provides valuable insights for future research.

- AI applications make significant contributions to the optimisation of energy systems. The literature highlights the substantial impact of AI on the efficiency improvements in solar energy systems. Consequently, the development and implementation of AI algorithms will enhance the overall performance of energy systems.
- Current literature indicates that AI possesses the capability to enhance the accuracy of energy demand forecasts. Therefore, integrating AI techniques in future studies is crucial for further improving energy demand predictions, especially considering the impact of climate change data.
- The integration of AI into hybrid energy systems promotes the collaborative use of energy resources. Further research in this area will enhance the energy reliability of these systems, contributing to the adoption of renewable energy solutions.
- The development of AI-based real-time monitoring systems will enhance system performance and expedite fault detection processes. This advancement will not only reduce maintenance costs but also provide advantages in improving the resilience of the systems.
- AI applications not only enhance the performance of solar energy systems but also have the potential to reduce carbon emissions. Therefore, the adoption of systems integrated with AI should be encouraged to ensure environmental sustainability.
- AI enables more efficient utilisation of energy resources. This not only provides environmental benefits but also assists in reducing costs through energy savings.
- Training energy professionals in AI contributes to the effective implementation of these technologies. Educational programmes are essential for enhancing workforce efficiency and encouraging the adoption of innovative solutions.
- Governments can promote the use of renewable energy sources by developing policies that support AI applications. Policymakers should establish regulatory frameworks at the national level and create incentive programmes to facilitate the integration of AI-based systems. This approach can accelerate the adoption of innovative AI applications by providing funding to support research and development activities. Furthermore, organising training programmes and workshops that encourage the implementation of AI technologies will ensure the cultivation of a skilled workforce in the energy sector.
- It should be emphasised that AI supports environmental sustainability. The integration of AI with solar energy systems plays a significant role in combating climate change by contributing to the resolution of environmental issues.
- Collaborations among stakeholders in the energy sector create opportunities for the development and implementation of AI applications. These partnerships can accelerate the integration and optimisation of energy systems.
- Awareness of AI and solar energy systems should be raised within the community, and public participation in their adoption should be encouraged. This effort can help facilitate broader acceptance of renewable energy solutions among larger audiences.
- Financial support mechanisms should be established for renewable energy projects. Providing the necessary resources for the development and implementation of AI-based systems will contribute to their widespread adoption.
- Long-term strategies should be developed and implemented for the integration of solar energy and AI systems. This will enable the enhancement of existing systems as well as the adoption of new technologies.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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